

Introduction

- 2-3 bullet points talking about the background for the project, previous similar projects, prior art, etc.

Purpose of the project, including final goal and timeframe

Info Panel

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Main Panel

- Explanation of the entire system/design.
- This panel often centers on a picture or diagram of the whole project and serves to point out relevant components/sections

Specific Guidelines

- Pictures & diagrams are worth 1000 words - use them instead of large blocks of text wherever possible
 - People will rarely read large amounts of text at showcase/conference - tell them the details verbally and let them look at pretty pictures
 - Focus on the really important & impressive things about each part of the project you want people to remember
- Can mix & resize sections any way - this is just a layout we used on a previous project
 - Don't change the size of the slide, this is what works with the poster plotter & blue boards
- Keep poster background and border colors to white, grey and blue to help color images stand out
- Keep general fonts & format the same as template to keep posters visually similar across projects
- Tables are a great way to organize top-level data about a project
 - eg: rocket apogee, total impulse, recovery system type, diameter or whatever else you think is important

What is Eclipse?

Rice's rocketry team! We develop launch vehicles for intercollegiate rocketry competitions, rocket engines, avionics hardware & software!

Info Panel

- Each info panel includes data or details about a specific part of the project
- Include pictures, diagrams, tables, timelines, etc especially if they are visually appealing

Future Plans

- Optional, but helps ground the project
- Timeline on how the project will be used in the future, what it's leading to, etc.

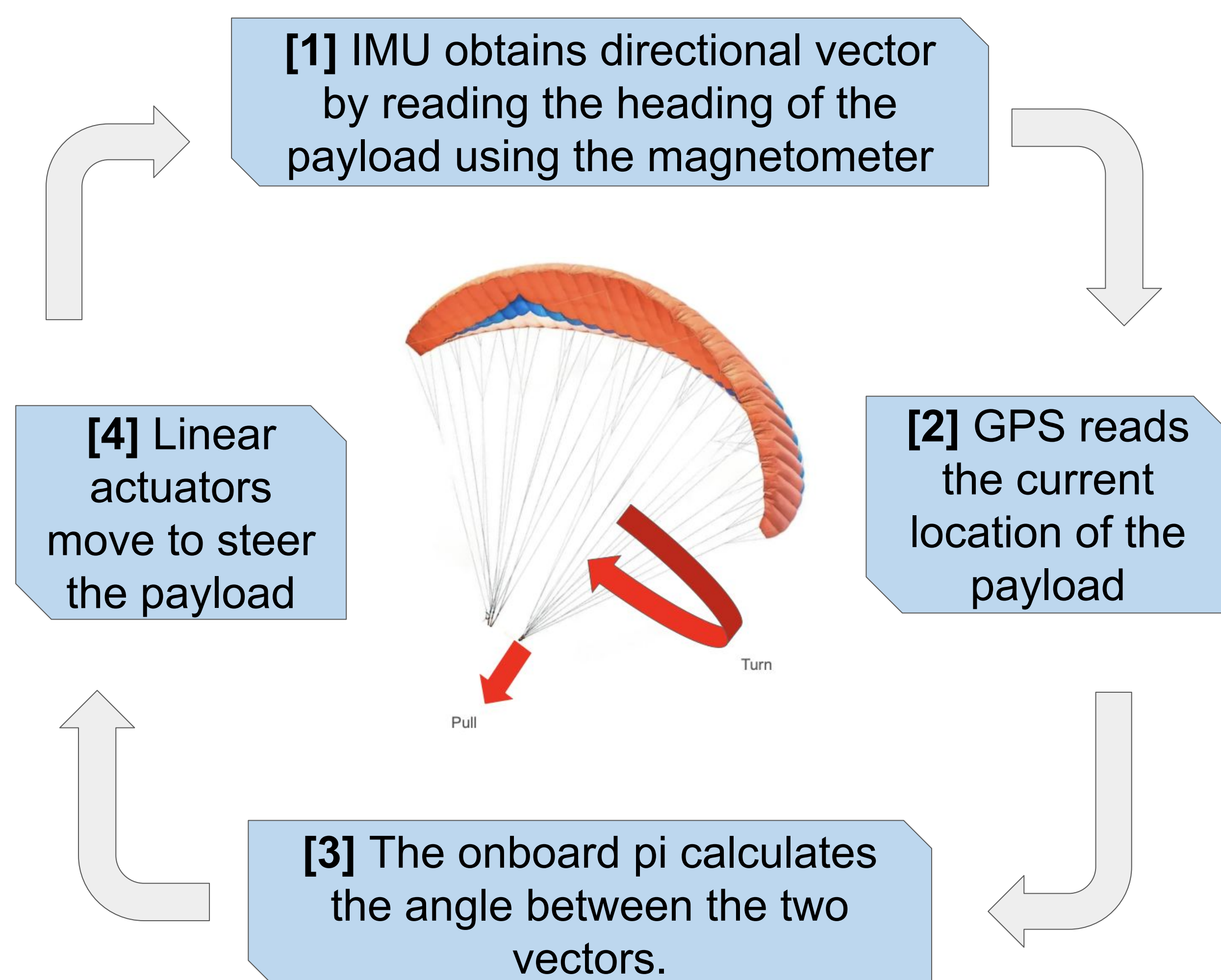
Steering a Payload

- Uses real-time data to **navigate to a target location**
- The payload is **released at 3,000 feet in the air** during a rocket's descent

Project Goal

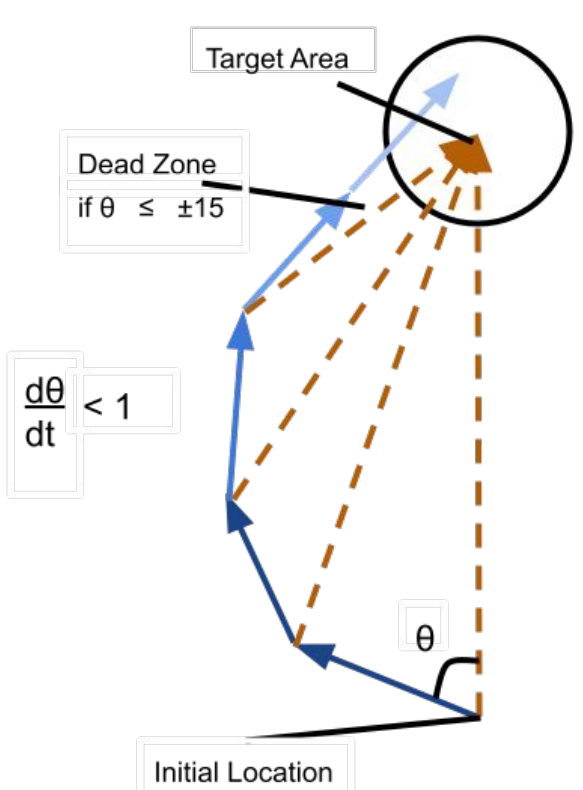
To design a system of **automatic recovery** for a **payload** inside a rocket, providing a basis for future **autonomous full-rocket recovery**

Navigation



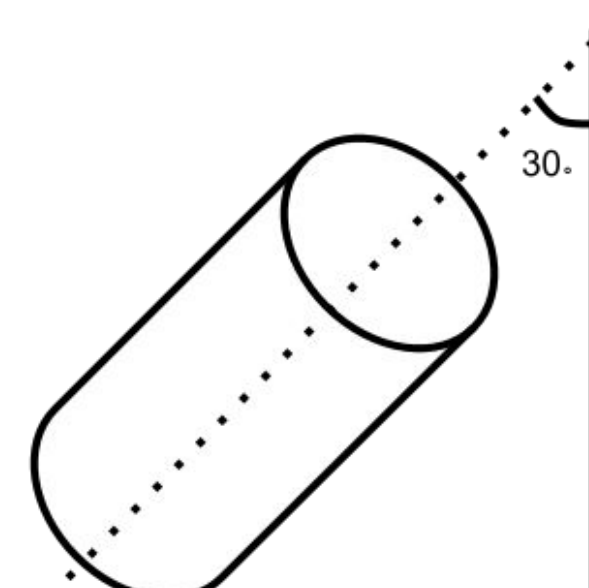
Deadzone

If the payload is within 15 degrees of the target vector, the payload will continue in the direction it is going to prevent overcorrection



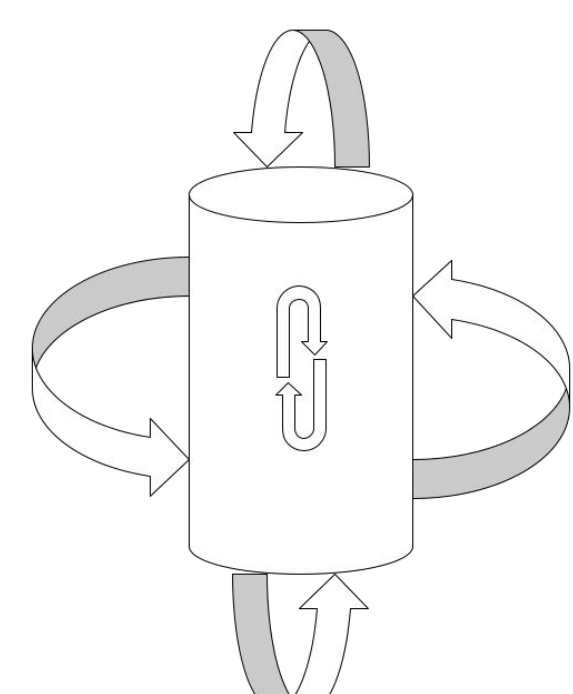
Tilt Correction

If the payload is tilted over 30 degrees from the vertical axis, data will not be collected, to prevent errors due to stability

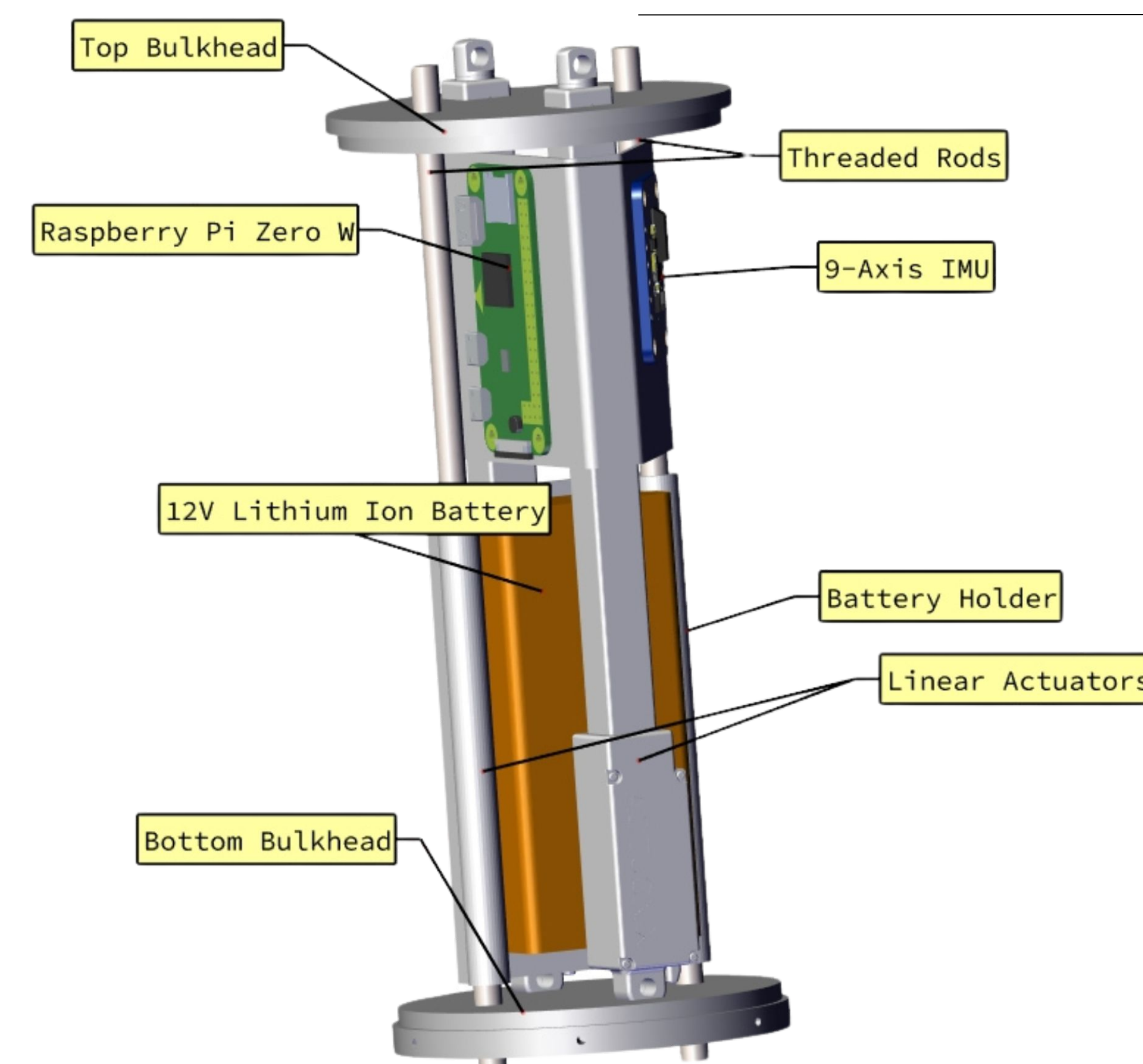


Calibration

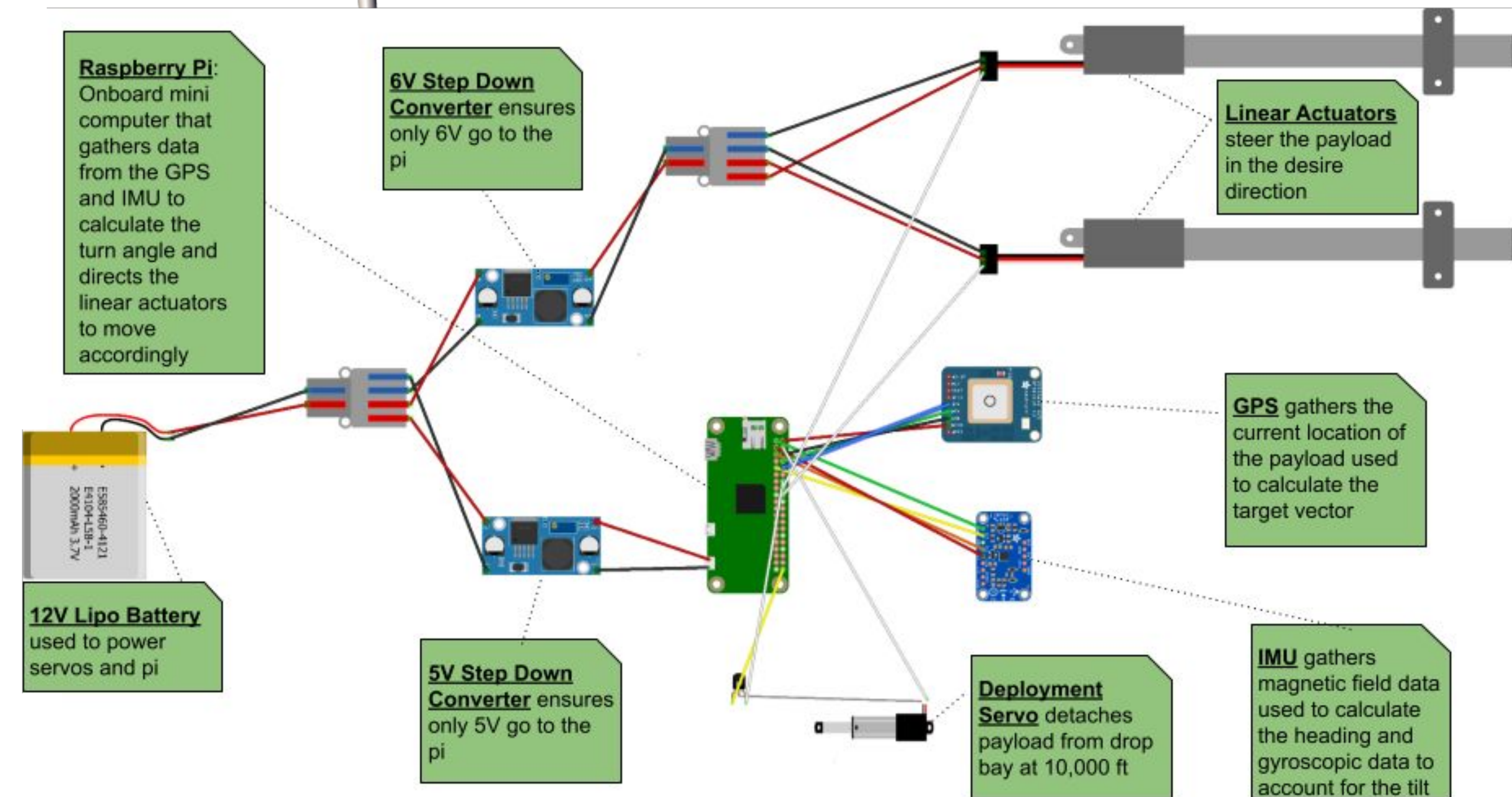
Software allows for automatic calibration of the IMU to correct for bias



Overall Payload Design



- ✓ Fiberglass bulkheads ensure **radio frequency transparency** and **strong support**
- ✓ Threaded Rods and linear actuators provide **vertical reinforcement**
- ✓ Electronics attached to outer perimeter of **3D-printed sled** so wires do not block RF communication
- ✓ **12 Volt 2200mAh LiPo battery** attached to PLA bracket with **velcro**
- ✓ Quick and easy assembly



Failsafe

In the event of navigation failure, the payload enters its failsafe protocol and pulls on one end of the parachute causing it to spiral downward

Drone Testing System

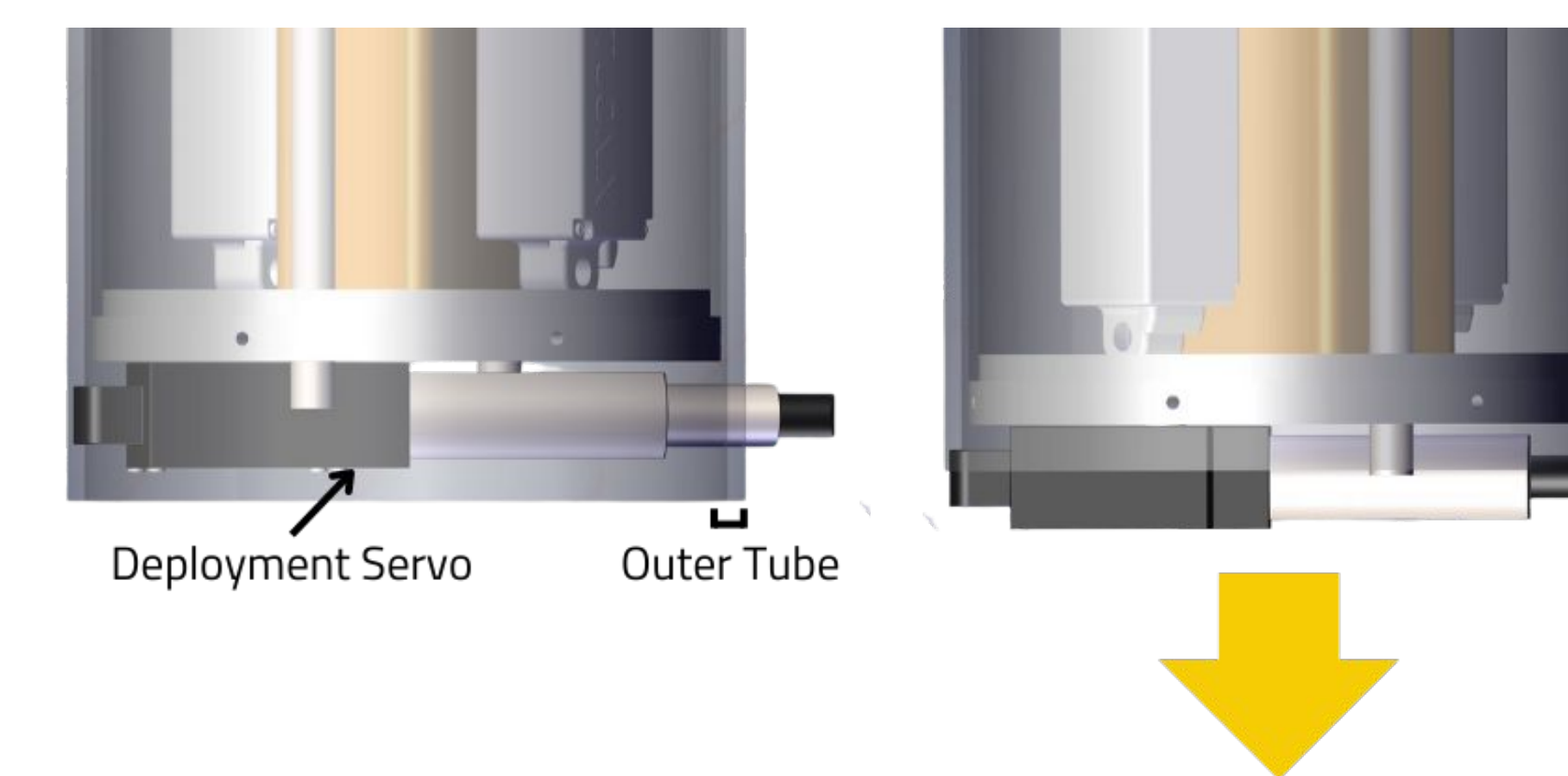
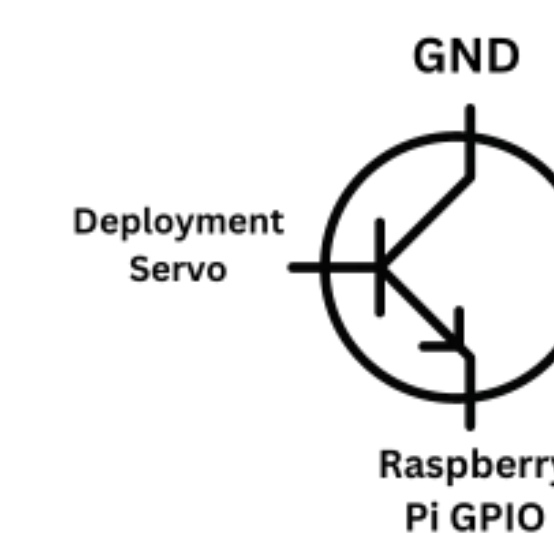
Phase 1



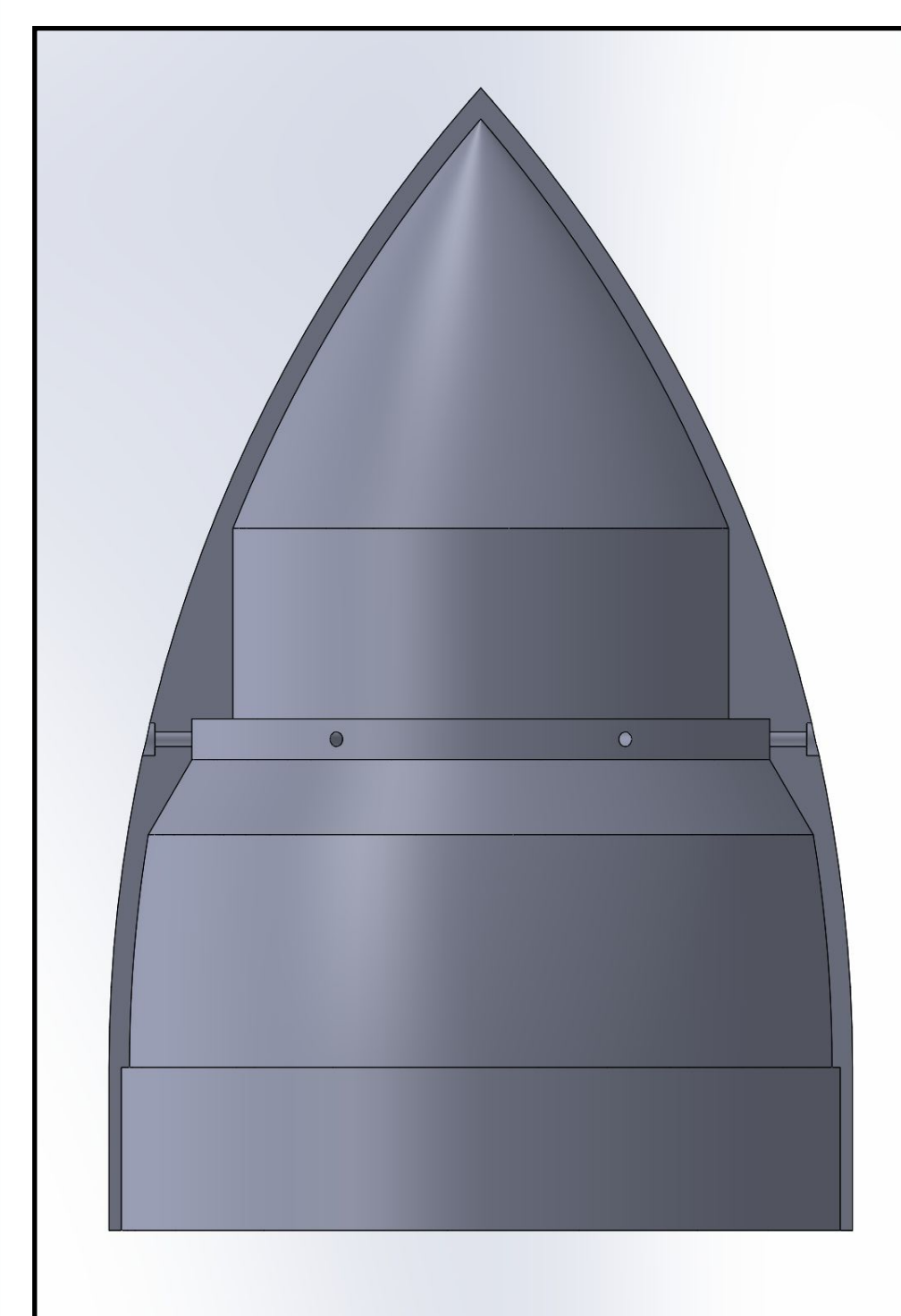
A **PLA Outer Tube** encases the payload. The **Deployment Servo** is left extended. The **carbon-fiber-framed drone** attaches to the top of the Outer Tube, flying **400 feet** in the air.

Phase 2

At 400 feet, a **GPIO pin** on the **Raspberry pi** sends a signal to the emitter pin on the transistor causing the servo to **contract**. The entire payload navigates itself to the desired location.

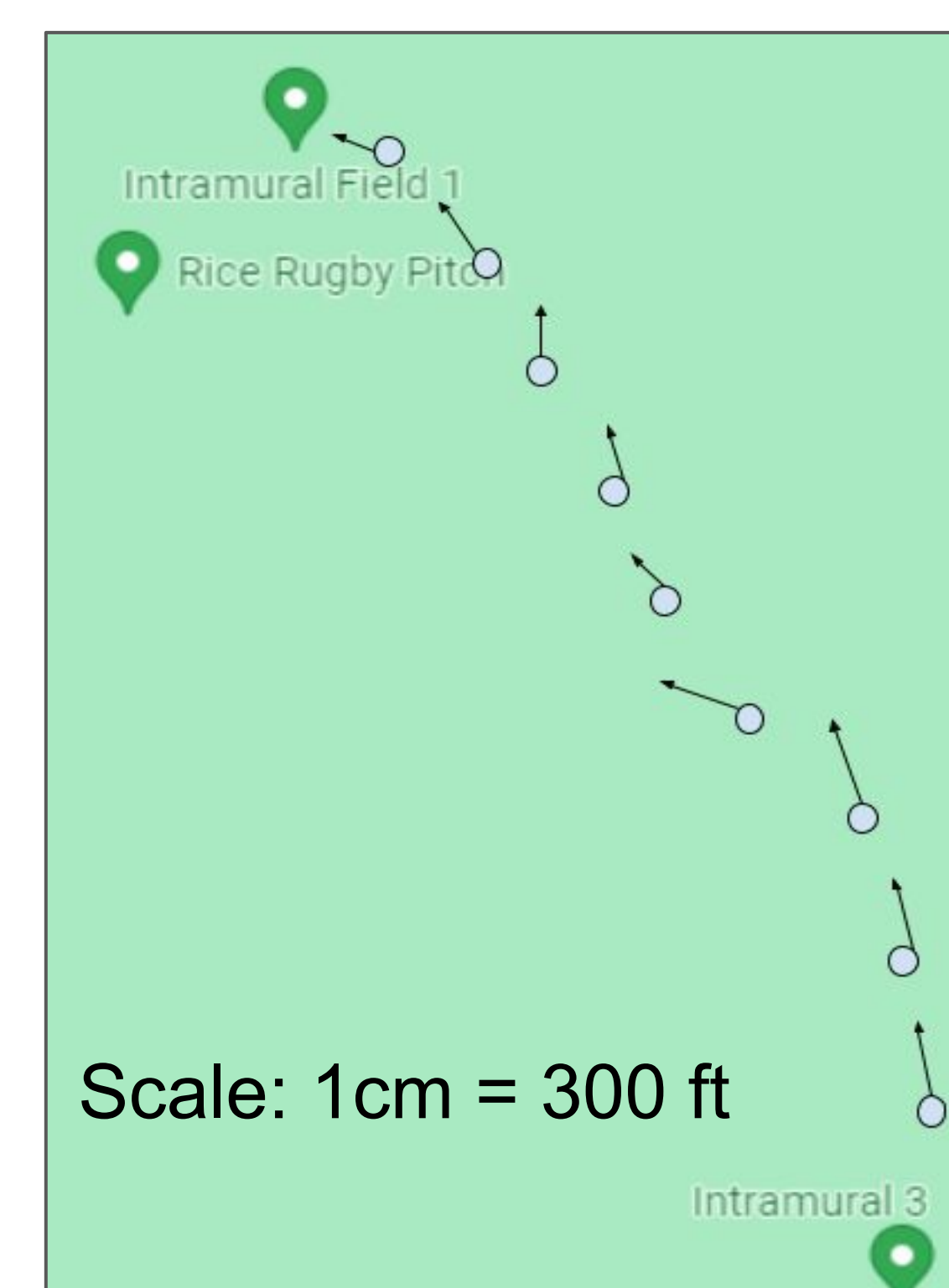


Nose Cone



- Korekau is attached to the Nose cone of the launch rocket.
- Nose cone deploys with Korekau to protect the payload.

Results



- Payload starts at intramural field 3 in the bottom right
- The payload successfully navigates to intramural field 1 in the top left
- Covers a distance of over 0.5 miles over a 400 ft change in altitude

Future Plans

- Deploy the payload from a high-powered rocket a second time to ensure correctness of design and algorithm
- Apply the system to a full-scale rocket payload, in order to enable fully autonomous recovery of rocket payloads

Introduction

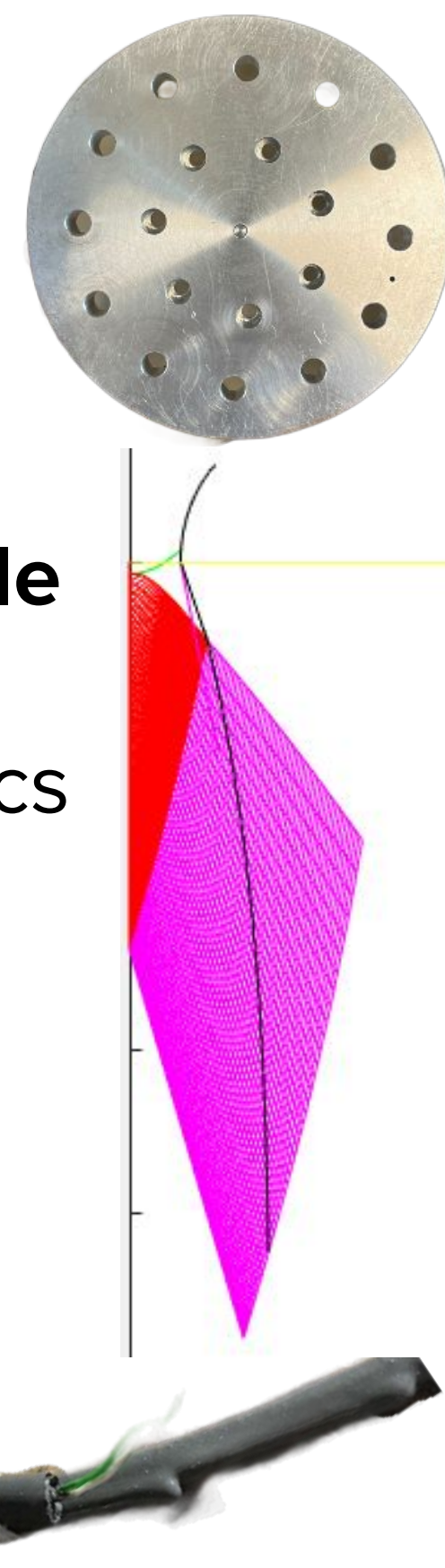
- We develop launch vehicles, rocket engines, and avionics for **international rocketry competitions**
- For the past decade the team been working towards flying a **completely student designed rocket with a hybrid engine**

Project Goal

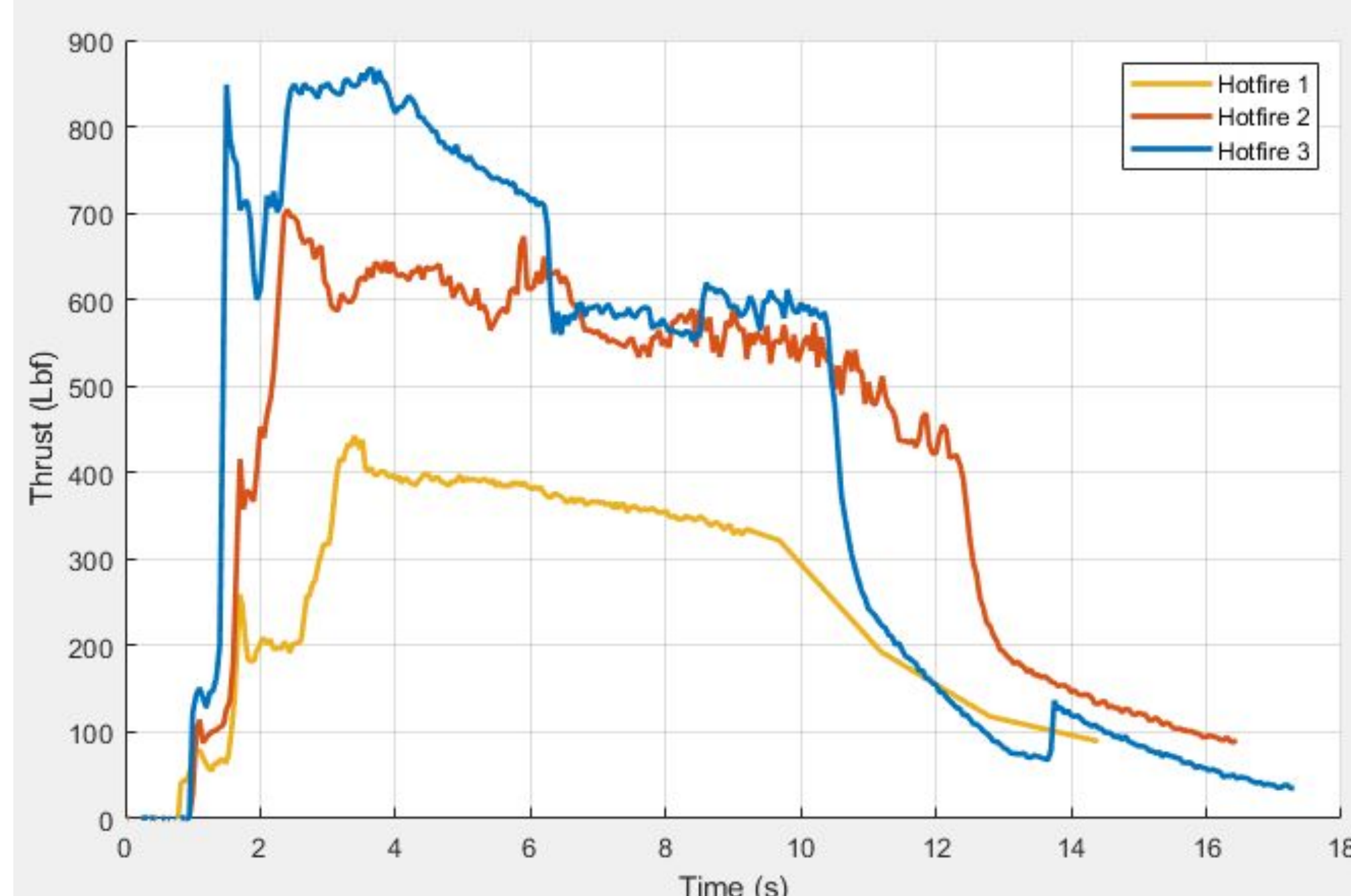
Design, build, and launch a **custom rocket airframe** propelled by a **student-built hybrid engine** to **20,000 feet**.

Propulsion Improvements

- Oxidizer Injection Method**
 - x7 Injector Hole Shower Head
 - Achieves desired injector hole area for proper mass flow rate of nitrous oxide
- Method of Characteristics Nozzle**
 - Geometry is numerically optimized to flow-characteristics
 - First tested MoC Nozzle
- Powerful Ignition Method**
 - Increases heat absorptivity
 - Pyrogenic mixture made of potassium nitrate, magnesium powder and epoxy
- Performance**
 - 1186 lbf Peak Thrust (+170%)
 - 663 lbf Average Thrust (+65%)
 - 7104 lb-sec Total Impulse (+3%)



Hot Fire Thrust Data



Full Archimedes Assembly

Engine Assembly

Specifications

Metric	Value
Rocket Length	16.8 ft
Velocity Off Rail	86 ft/s
Peak Motor Thrust	1,186 lbf
Total Impulse	7,099 lb-s
Peak Altitude	22,701 ft
Time to Apogee	40 s
Max Velocity	1194 ft/s (Mach 1.12)

Oxidizer Fuel Tank

Aluminum tank that holds 40 lbs of Nitrous Oxide at 750 psi



Combustion Chamber

Aluminum chamber with a phenolic lining and HTPB fuel grain

Fins

Constructed with Nomex® Honeycomb core & layered carbon fiber

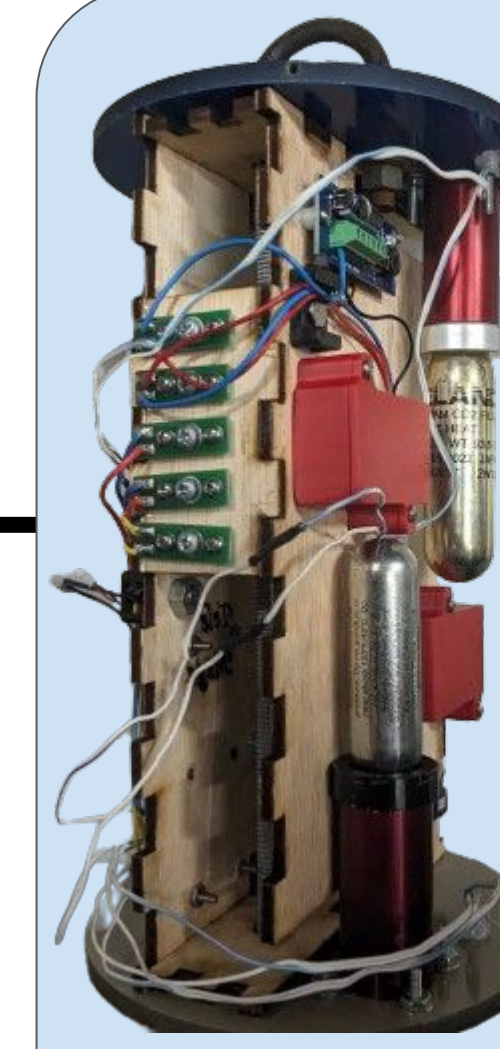
Boat Tail

Constructed in-house from carbon-fiber to reduce drag

Launch Vehicle

Nose Cone

36.5" Von Karman Design for efficient aerodynamic performance
Houses the backup GPS



Avionic Bay

Controls parachute deployment from flight computer readings
Transmits GPS data via telemetry

Scientific Payload

3U Aluminum/Steel CubeSat that holds Arduino, piezo sensors, and accelerometer to record vibration data

Camera Block

Three 4K cameras mounted to a 3D printed piece record 360° video



Experiment

Vibration data and video used to simulate the experience of flying inside the rocket via haptics and VR

Engine Extension

Machined in-house from aluminum to couple the airframe and engine

In-House Composites

- Carbon fiber tip-to-tip fin reinforcement
- Strength validated via material testing
- High thermal resistance to extreme engine temperatures
- Body tube reinforcement minimizes fractures from stress concentrations



Custom Launch Rail

- Interfaces with modified trailer
- 40 ft radio tower with 1515 rail
- Raised with two electric winches
 - Cables routed through swivel pulley at top of steel bar
- Arrowhead anchors + Guy wires to ensure securement post-raise
- Load cell to monitor oxidizer filling



Eureka! Recovery Rocket

- Reached 12,000 feet
- Successful supersonic recovery
- System Testing for Archimedes:
 - Recovery configuration
 - Payload sensors and cameras
 - Barometric altimeters
 - CO₂ Separation Deployment
- Relaunching to collect more data



Future Plans

- Compete in New Mexico on June 17th**
- Test launch of Eureka! on April 13th
- Ground Separation Tests

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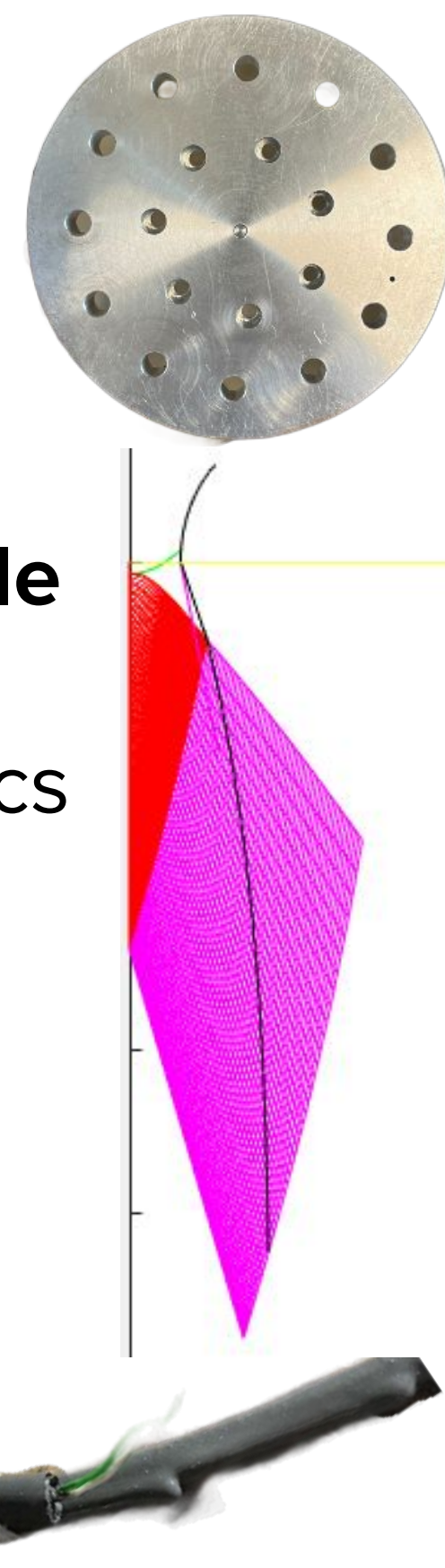
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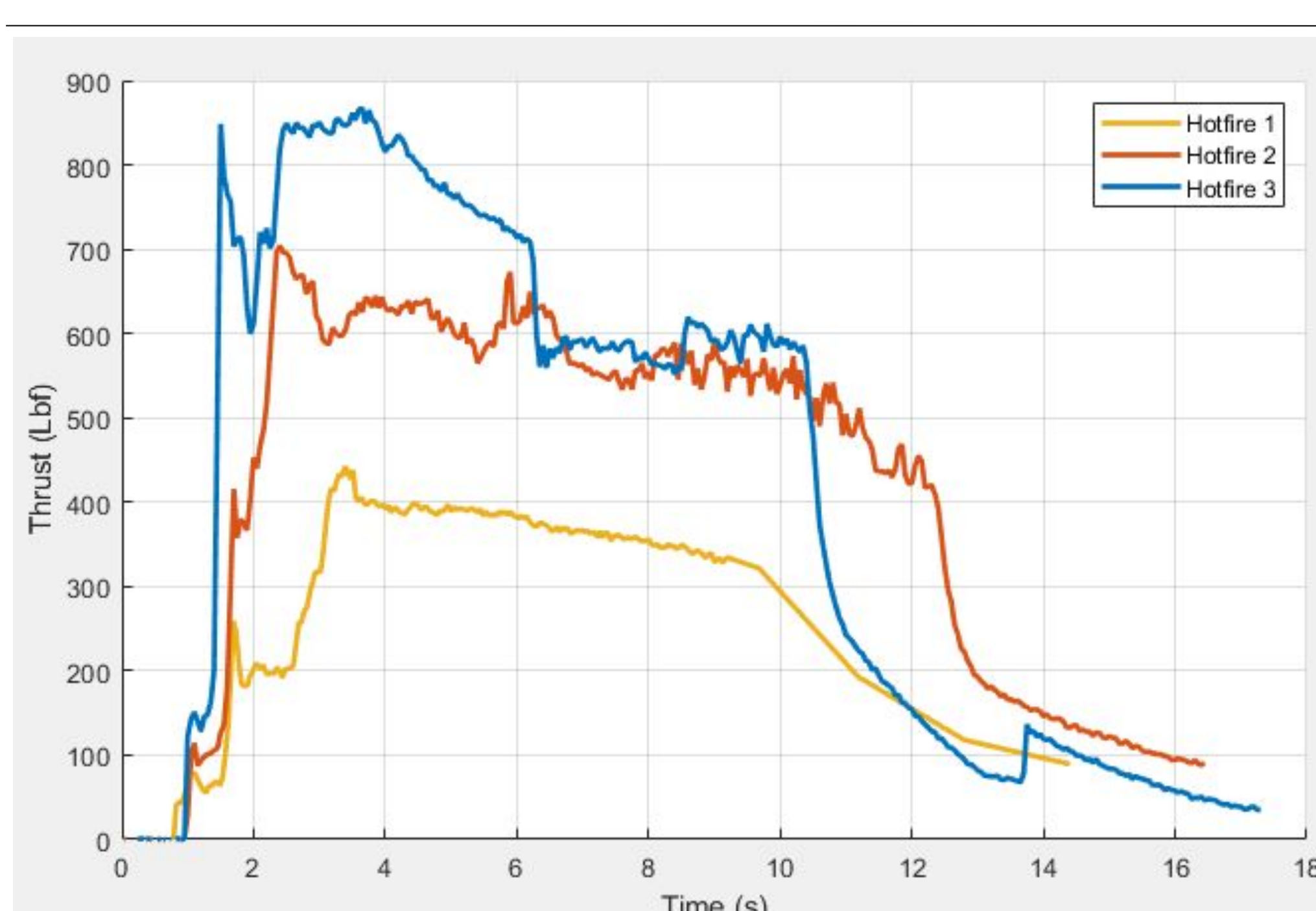
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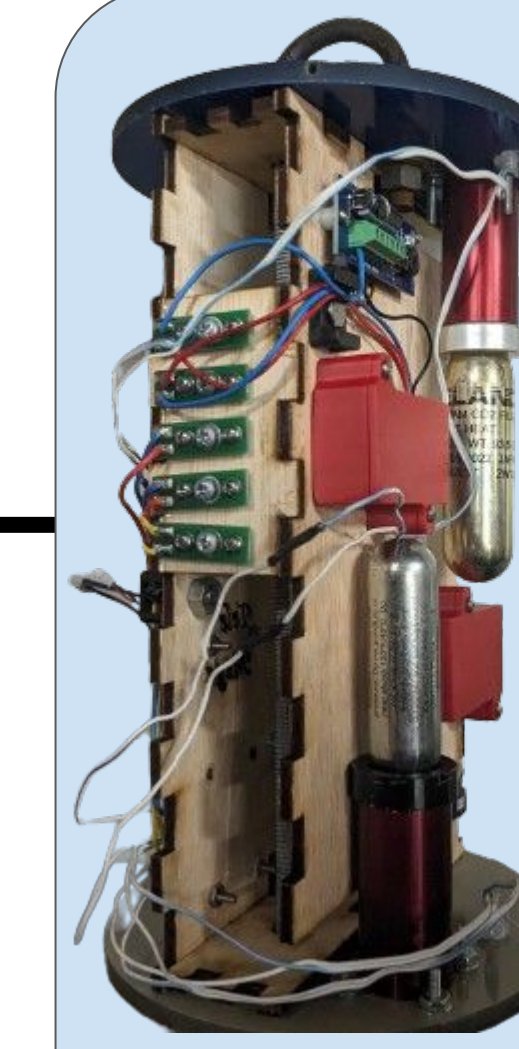
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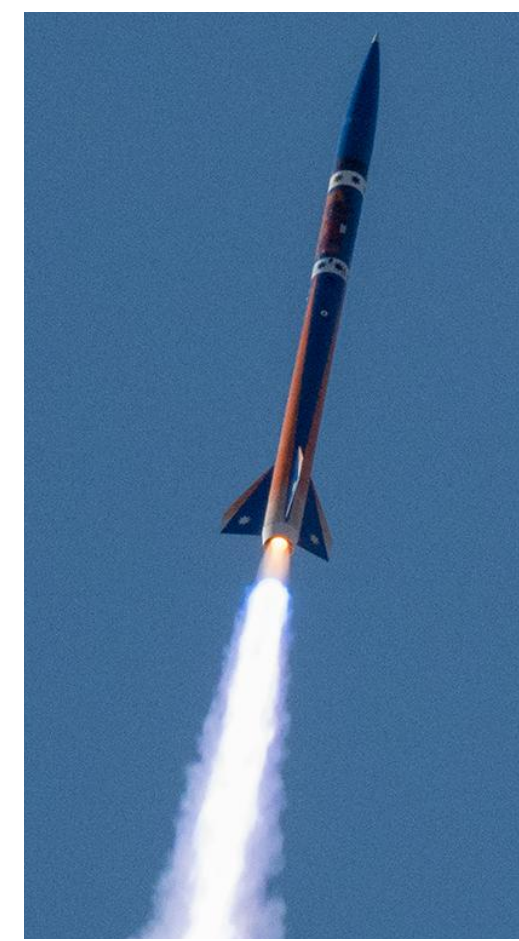
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- Custom boattail for improved aerodynamics

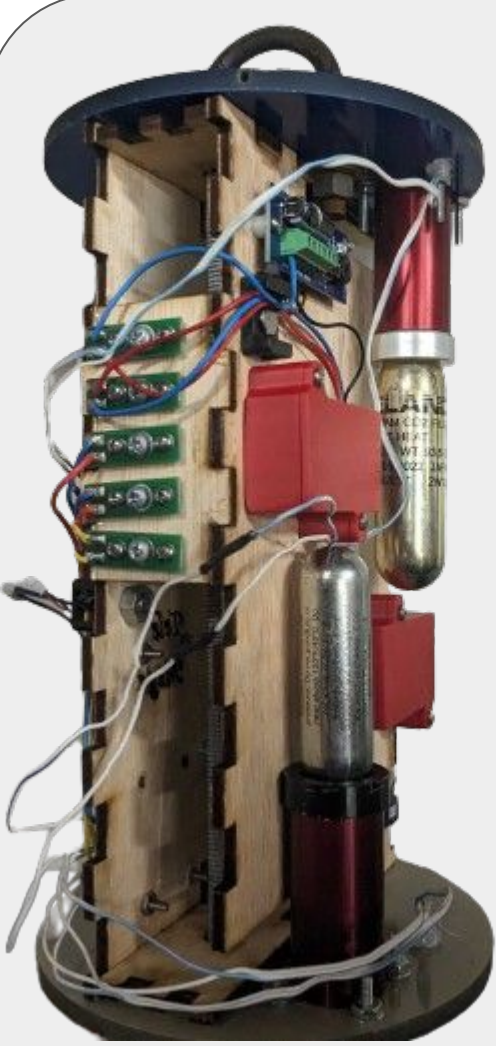
Better lower boattail pic, will get for final print

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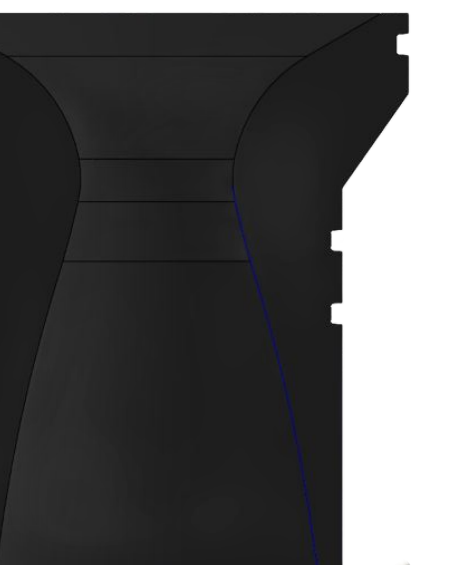
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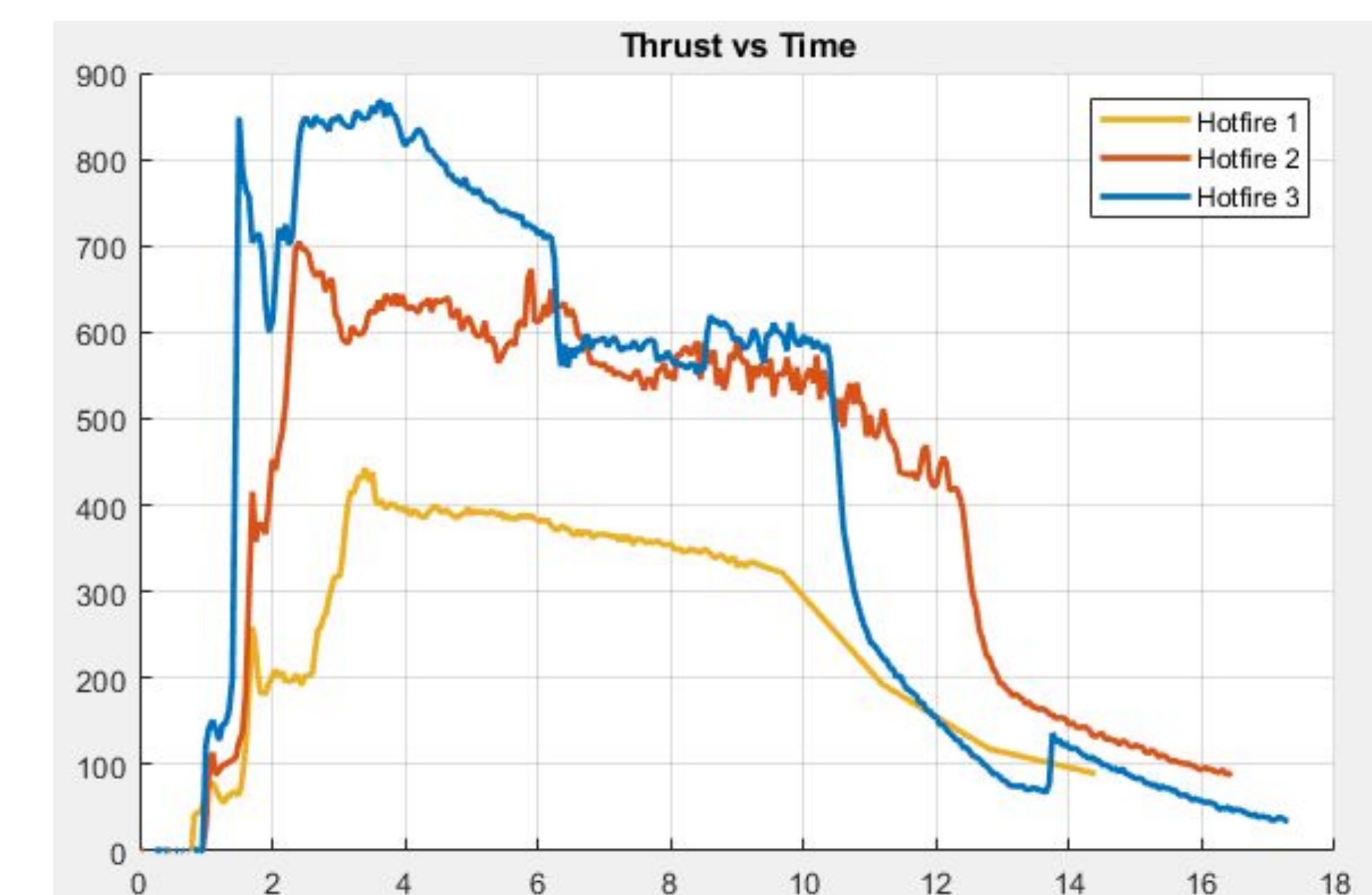
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Past Examples...

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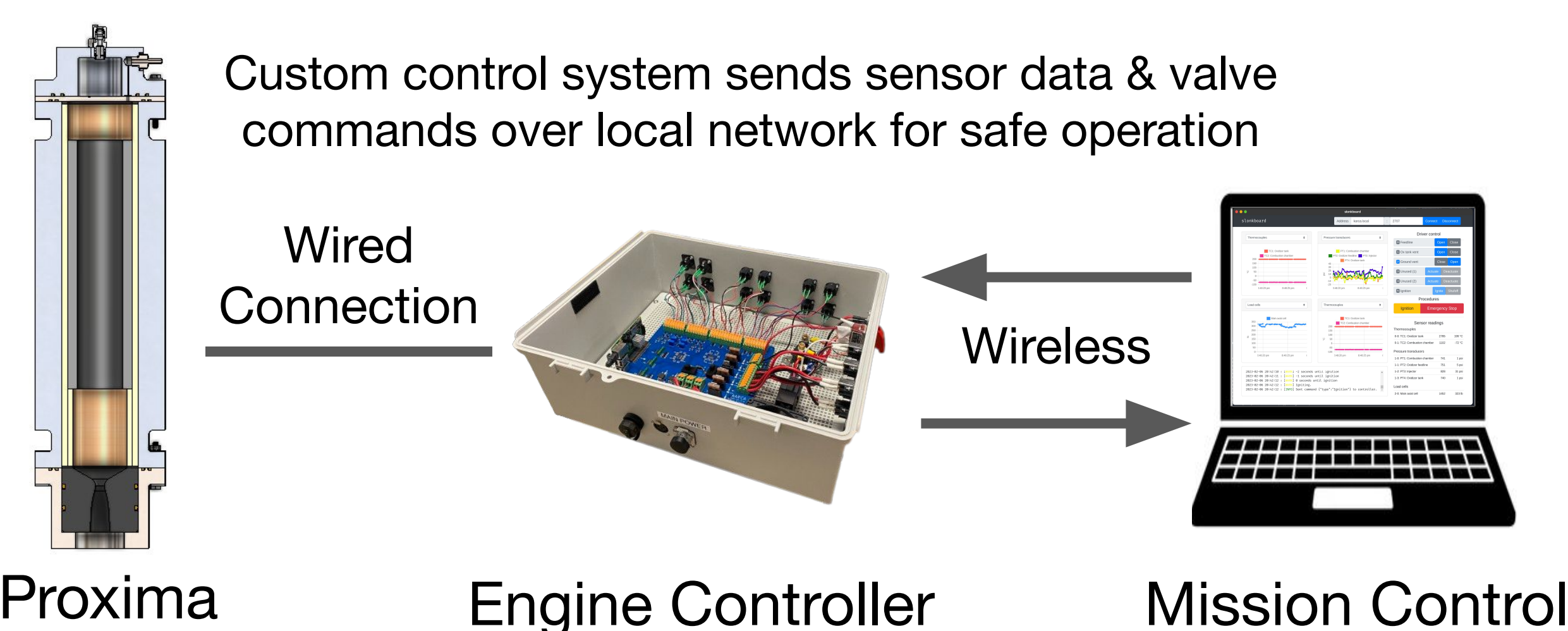
- Produces thrust with liquid oxidizer and solid fuel
- 10:1 scaled down version of Eclipse's first flight-optimized hybrid engine (Titan II)
- Adaptable research tool to develop club knowledge
- Apply innovations to design & testing to future hybrid systems

Design & build a new small hybrid rocket motor to test experimental components and improve testing procedures

Engine Characteristics

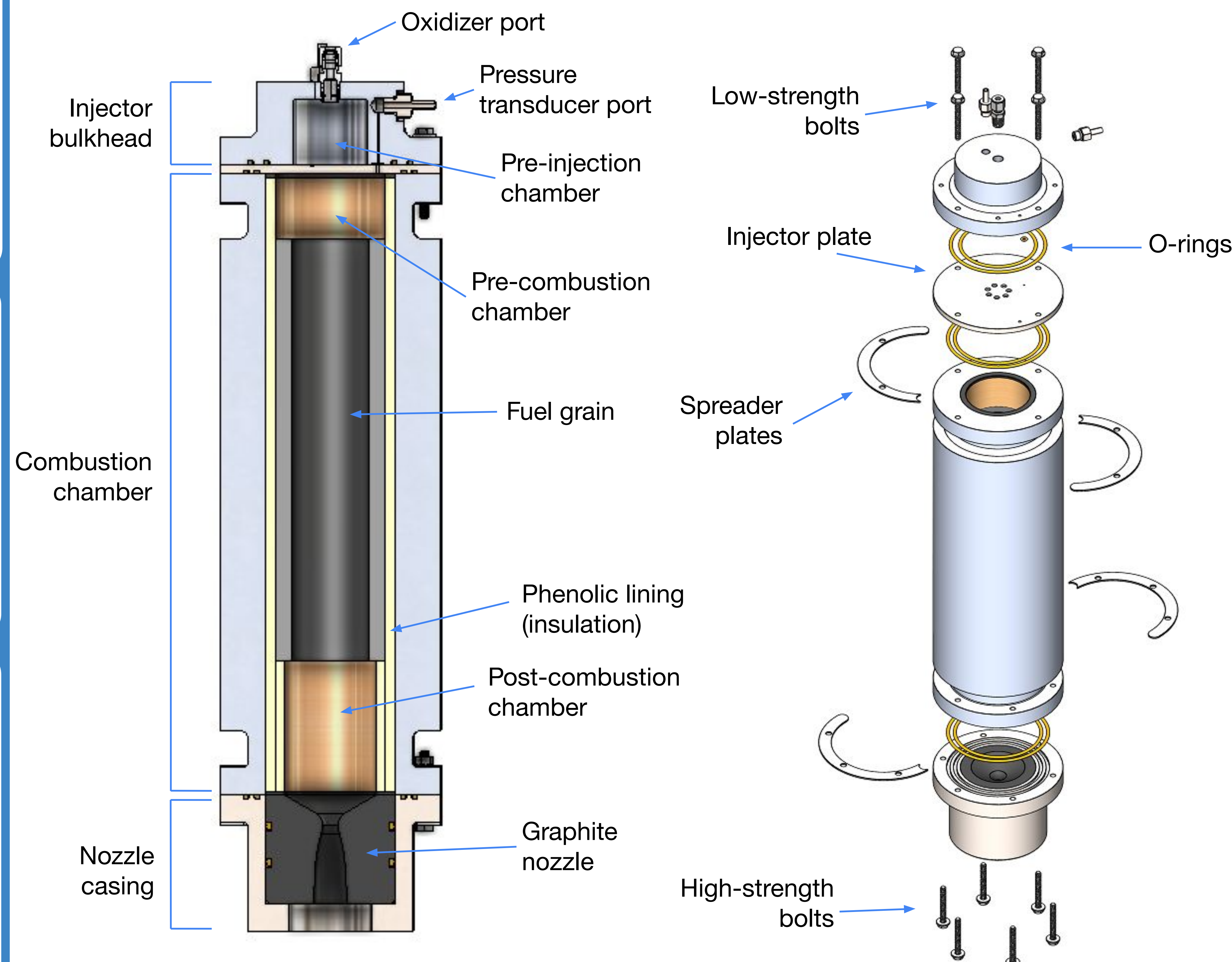
Thrust	92 lbf
Chamber Pressure	500 psi
Solid Fuel	Hydroxyl Terminated Polybutadiene (HTPB)
Liquid Oxidizer	Nitrous Oxide
Burn Time	10 seconds

Instrumentation & Control



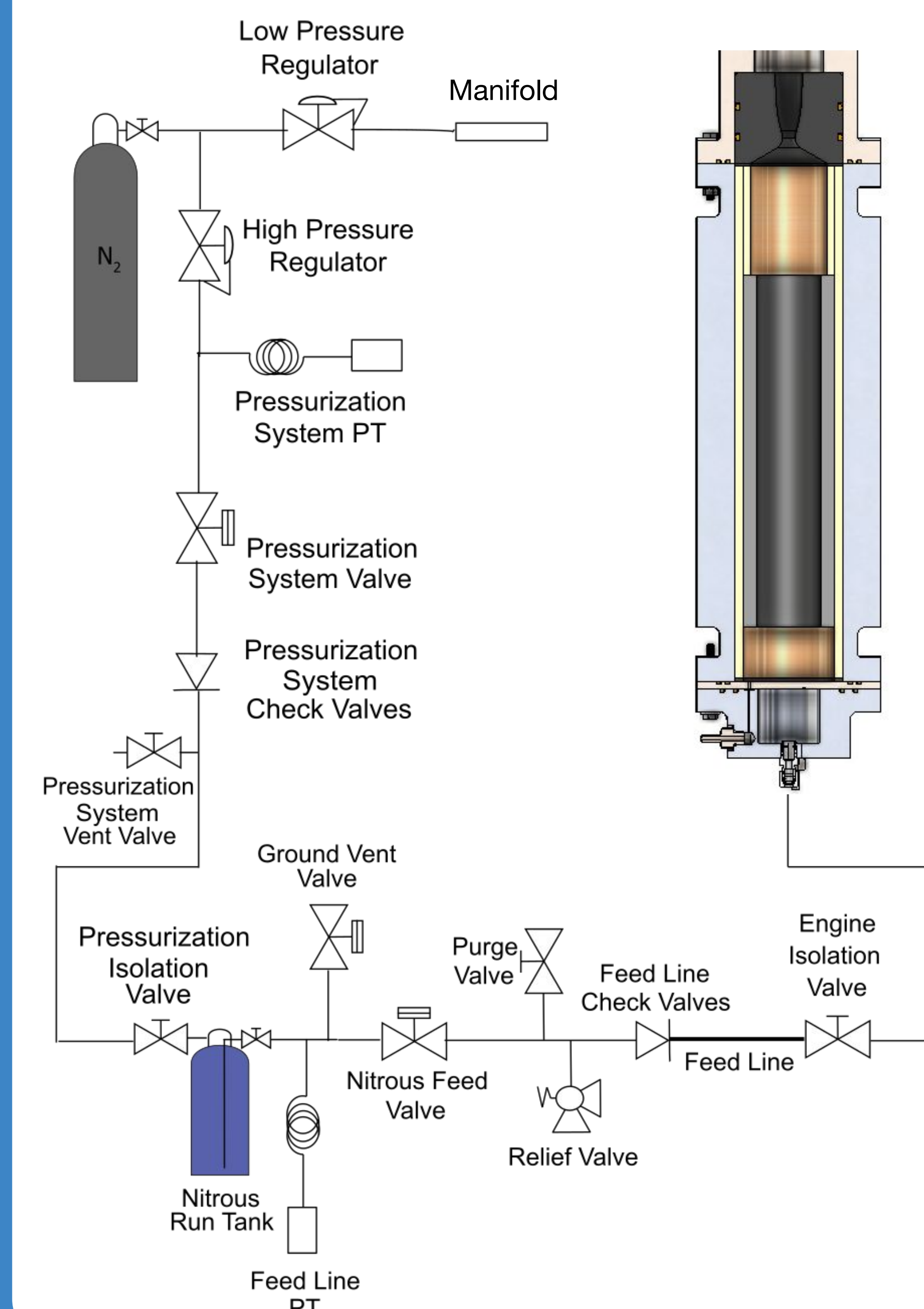
Overall Engine Design

- Designed for easy manufacturing/assembly & safe testing
- Components are easily replaceable for multiple engine tests per day with frequent test days
- Designed using lessons learned from previous engine



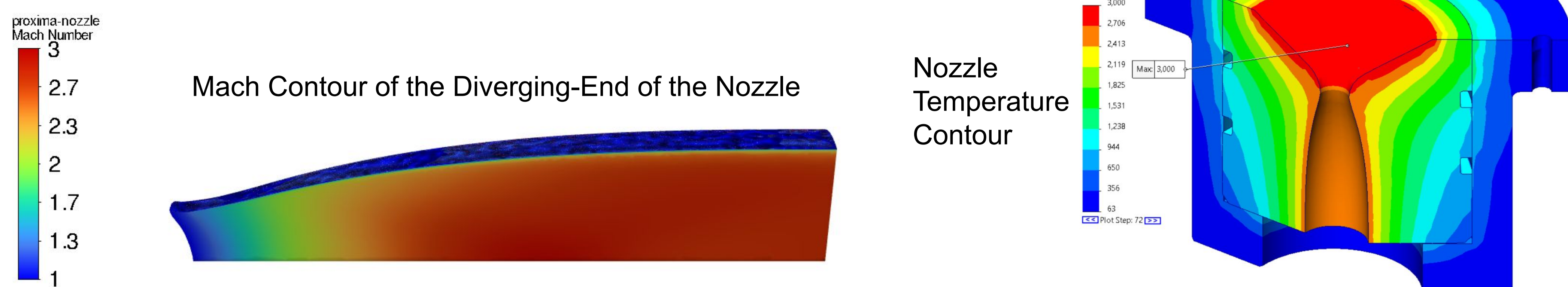
Plumbing System

- Oxidizer supply system & pressurization system maintain constant thrust throughout burn
- System designed to fail safely during overpressurization or power failure
- Simple construction for easy operation and fewer points of failure



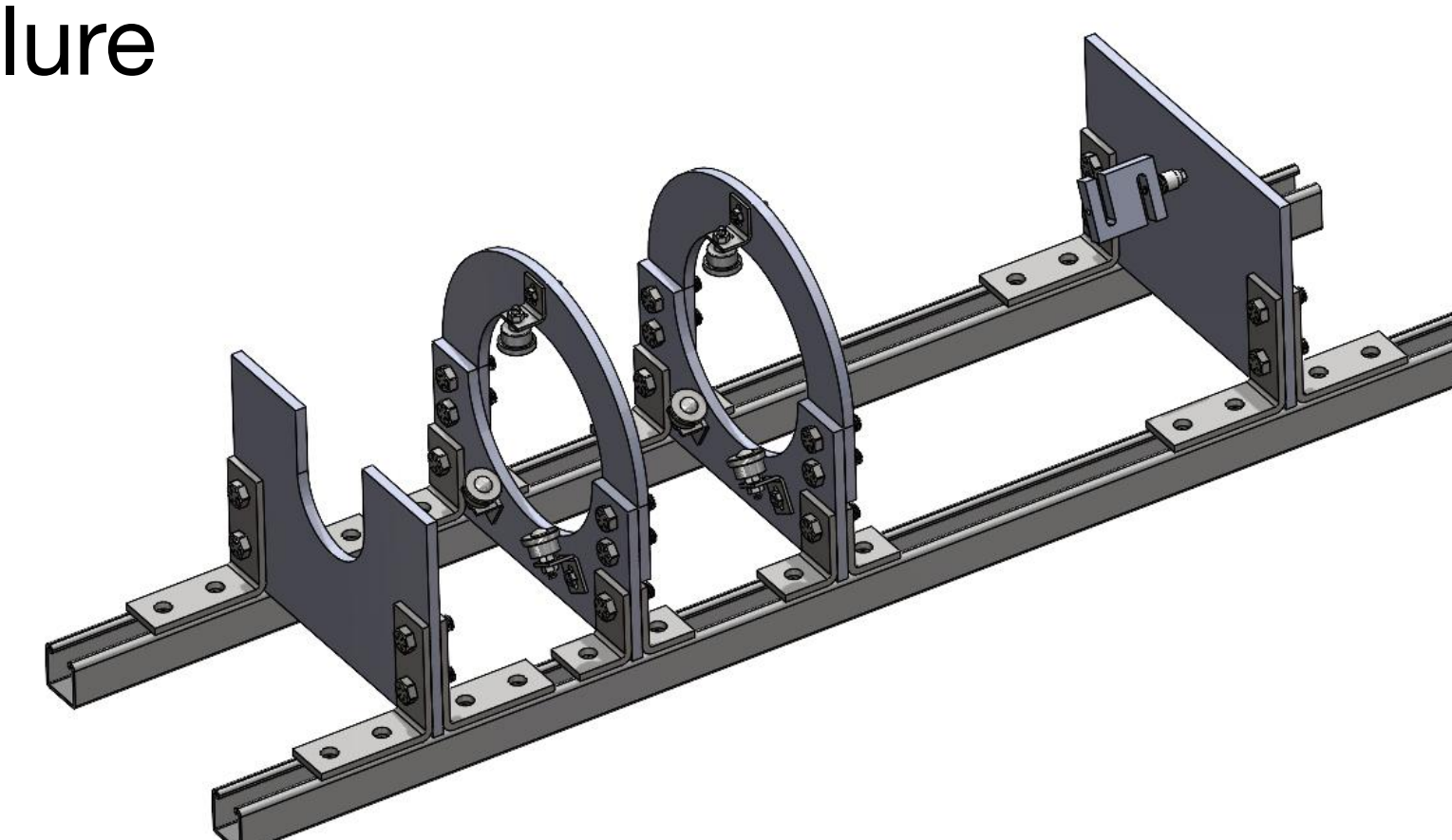
Simulation

- Performed extensive FEM stress analysis of all engine components & assembly to validate design choices
- Method of characteristics nozzle design code optimizes axisymmetric, curved mach line while minimizing length
- Transient thermal analysis of nozzle assembly to determine maximum safe burn time



Test Stand

- Simple, strong construction allows for easy assembly, transport and setup at test site
- Horizontal test stand eliminates flame diverter
- Integrated load cell for data collection
- Safely contains engine parts in case of failure



Future Plans

- Cold flow by end of Spring 2023 semester
- First hot fire September 2023
- Test engine components: experimental fuel grain compositions, injector plates & nozzles
- Use data & knowledge gained for future hybrid systems and liquid engines

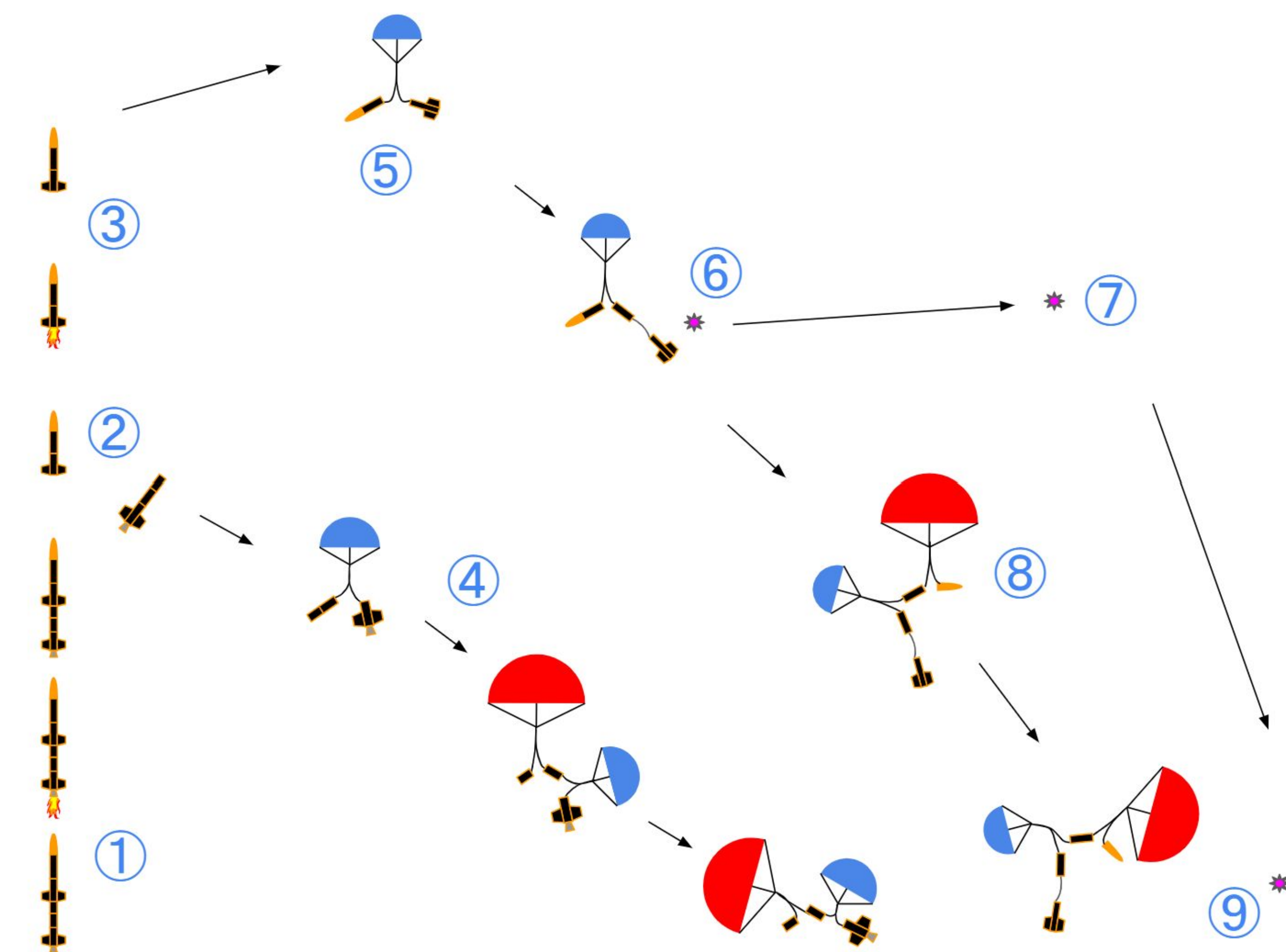
Introduction

- Rice Eclipse develops launch vehicles, rocket engines, and avionics hardware & software for the **Spaceport America Cup**
- The club's long term goal is to build a rocket to fly our custom hybrid engine, Titan II

Project Goal

Design, build, and launch a **custom dual-stage sounding rocket** to **30,000 feet** to deploy an **autonomous drone payload** for **ground mapping**.

Phases of Flight

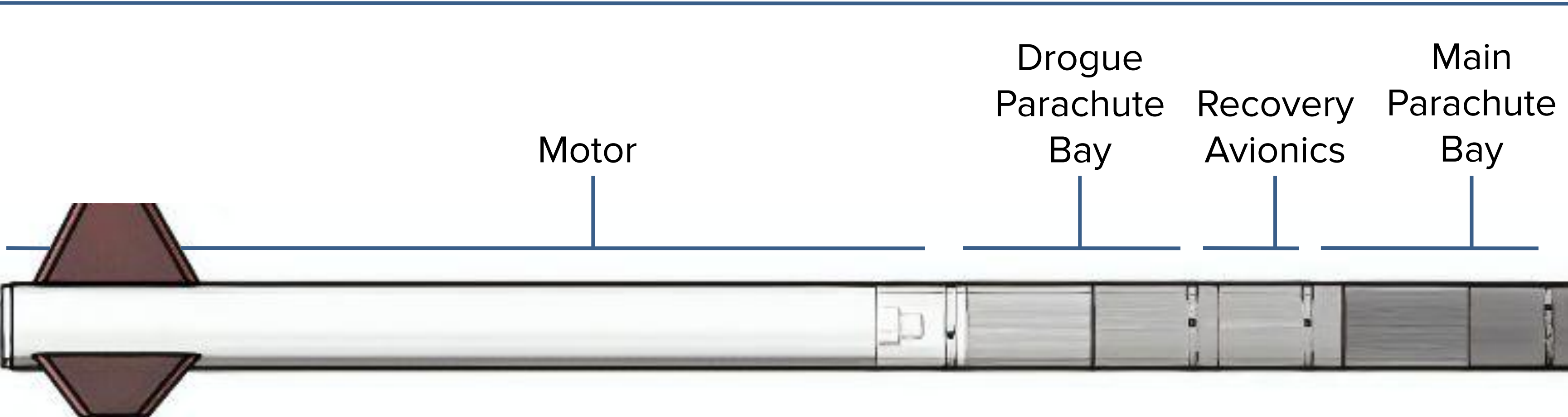


- ① **Booster Ignition:** $t = 0.00s$
- ② **Stage Separation:** $t = 4.59s$
- ③ **Sustainer Flight:** $t = 6.62s$
- ④ **Booster Main Recovery**
- ⑤ **Sustainer Apogee:** $t = 43.89s$
- ⑥ **Drone Deployment**
- ⑦ **Drone Flight and Descent**
- ⑧ **Sustainer Recovery**
- ⑨ **Drone Landing**

Rocket Specifications

Max Velocity	1913 ft/s (Mach 1.7)
Booster Peak Altitude	7,979 ft
Sustainer Peak Altitude	32,211 ft
Rocket Length	16' 4"
Weight at Liftoff	88.3 lb
Booster Motor Thrust	716 lbf
Sustainer Motor Thrust	454 lbf
Time to 30,000 ft	30.1 s
Max Acceleration	343 ft/s ²

Booster (1st stage): Dædalus



Carbon Fiber Body Tubes

Hand-rolled 7-layer carbon fiber body tubes

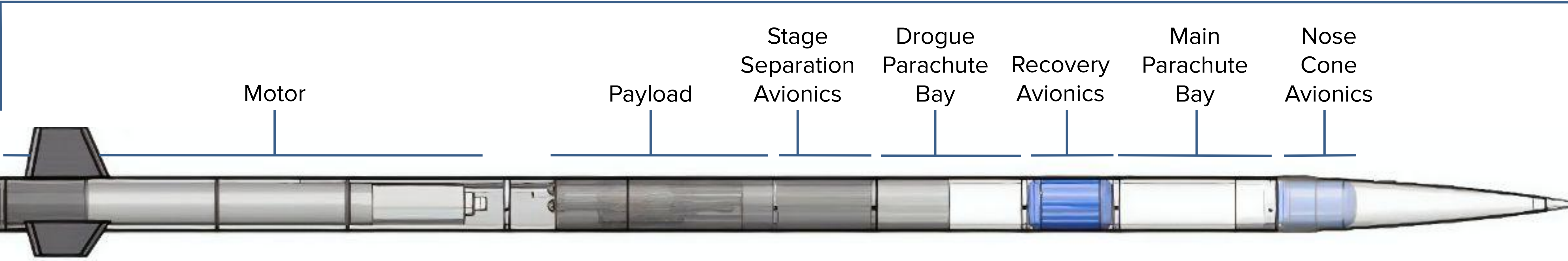
Compression, hoop stress, and three-point bend material tests to **establish flight durability** of components

Fins

Aramid honeycomb core & layered carbon fiber

Tapered fiberglass fin edges

Sustainer (2nd stage): Icarus



Lower Body Tube

Houses motor & payload

Motor & payload mounting points combined to reduce mass

Coupler

Fiberglass tube joins body tubes

Contains payload & stage separation flight computer

Upper Body Tube

Houses parachutes & recovery avionics

Parachutes anchor to U-bolt in aluminum bulkheads

Nose Cone

Fiberglass allows radio transmission

Contains GPS and custom flight computer

Recovery & Testing

- Dual-deployment system** for each stage
 - Apogee:** Deploy drogue parachute
 - 1000 ft:** Deploy main parachutes
- Altimeters** trigger **black powder charges** to separate rocket
- Successful recovery deployment** at 2.2K from Prestige flight
- Separation testing** of parachute chambers to ensure successful recovery



Separation Testing



Prestige

Payload: Experimental Geo-Mapping Drone

- Arms unfold** upon ejection at 3,000 ft via elastic bands
- Pixhawk** flight controller with custom ArduPilot software stabilizes flight and handles autonomy
- Raspberry Pi** signals Pixhawk to fly upon unfold using limit switch data
- Camera** takes periodic photos, stitched into a map after retrieval



Future Plans

- Compete in New Mexico at the **Spaceport America Cup 2023**
- Build Archimedes, our first rocket propelled to 30,000 ft by our **SRAD hybrid engine**
- Build our first **three-stage Spaceshot** rocket

Introduction

Rice Eclipse's hybrid rocket engines need an **integrated mission control system** which can support **real-time data acquisition and telemetry** as well as **long-range remote control**.

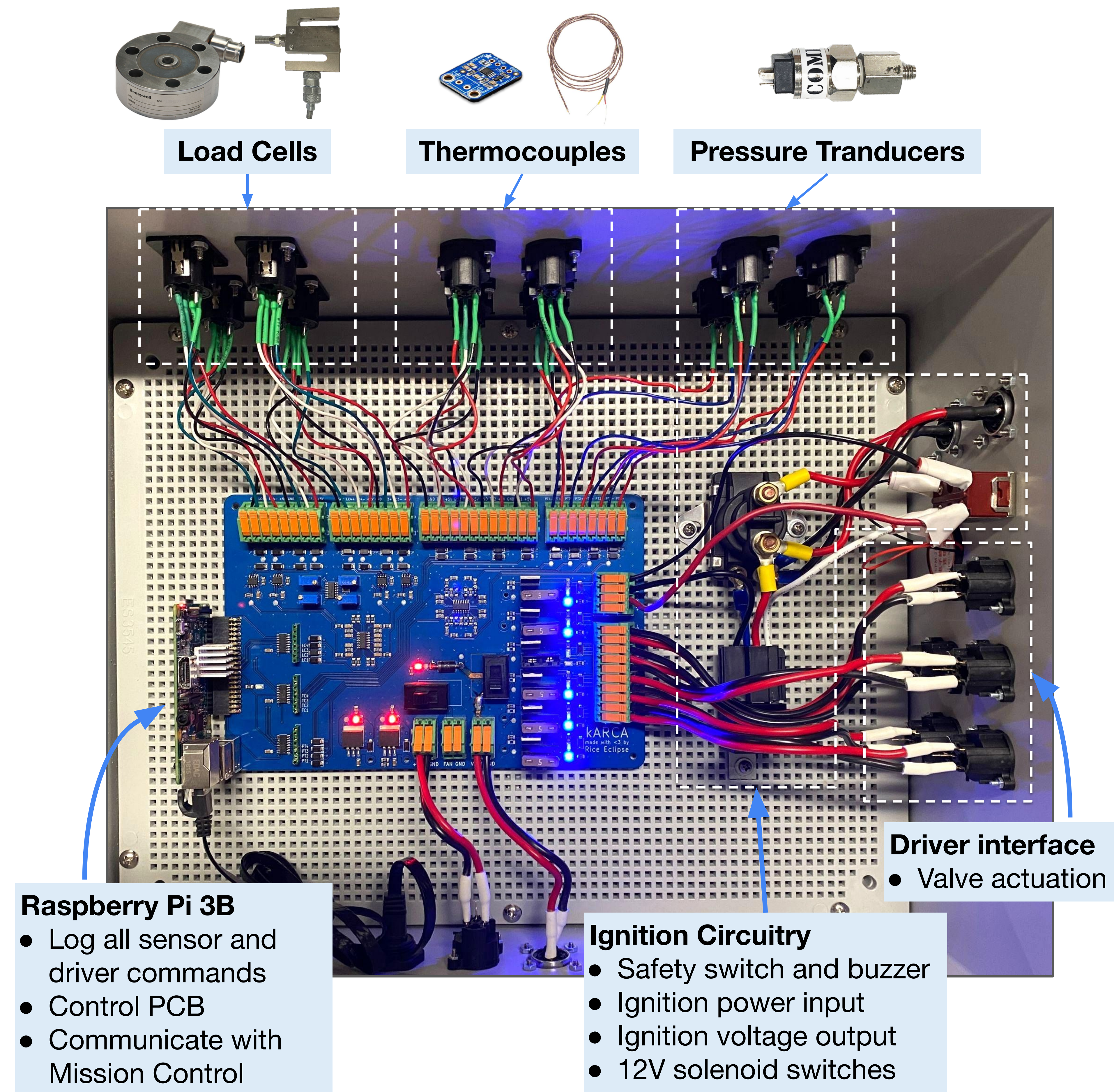
Project Objectives

- Custom configurable engine controller
- Adaptable for a wide range of engine designs with no changes to the hardware
- Dashboard desktop interface which allows the users to remotely control and observe the engine's state

System Specifications

- ✓ Control 5 solenoid valves
- ✓ Supply 50A ignition current
- ✓ Detect anomalies and automatically execute emergency shutdown
- ✓ Record pressure, force, and temperature data across 12 sensors
- ✓ Transmit data to Mission Control in real-time
- ✓ Maintain remote connection at least half a mile from test site (2460 feet)

System Outline

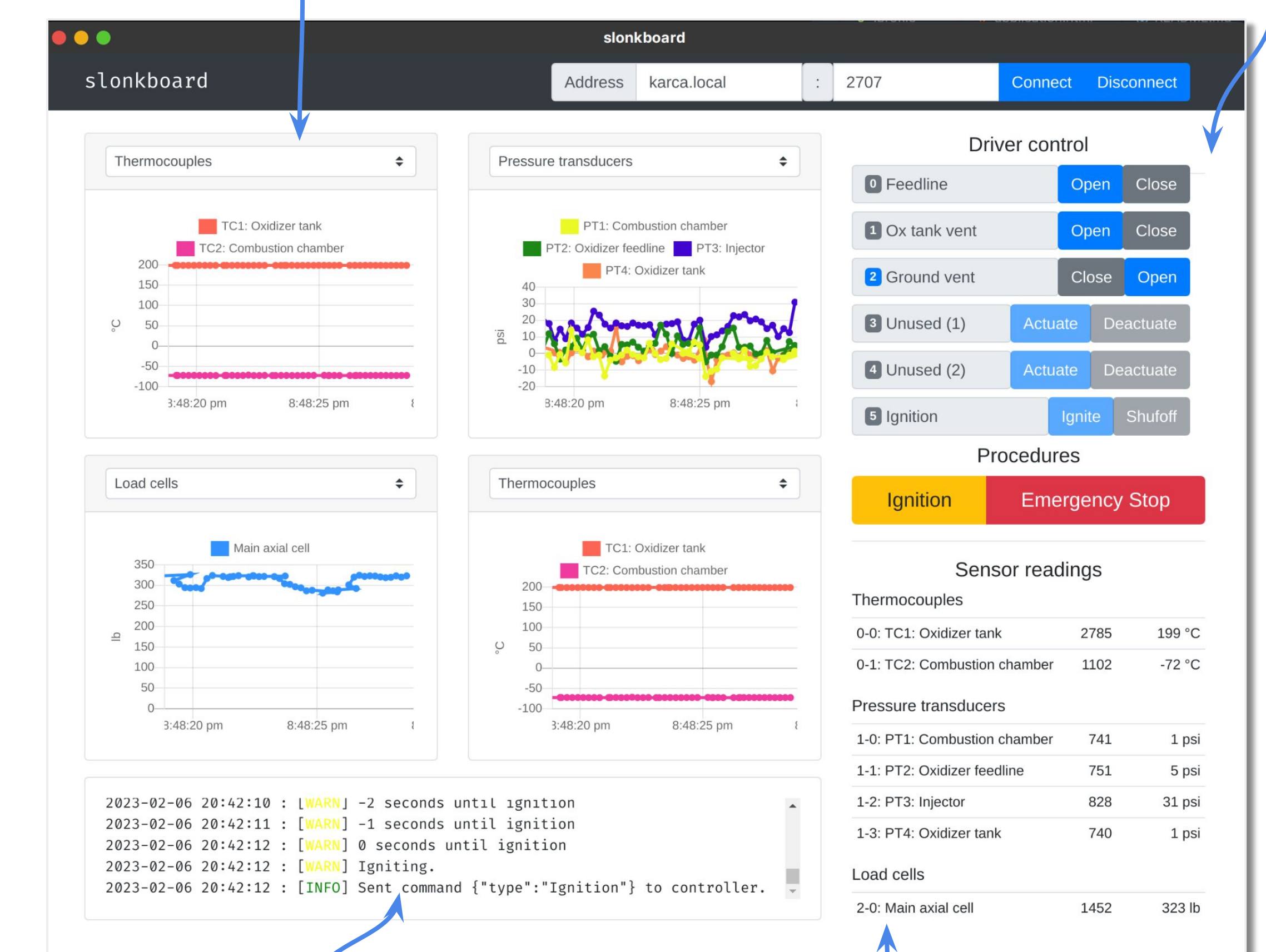


Real-Time Dashboard

Electron application with Node.js framework and npm package manager to handle related graphing, UI, and logging libraries.

Live data charts

Control panel

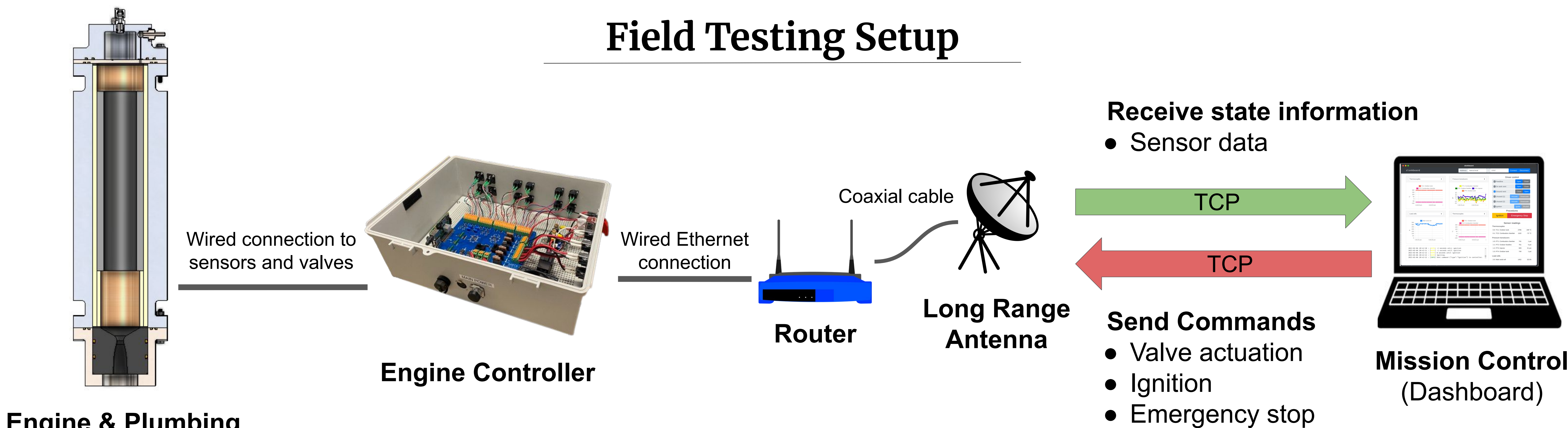


Event log

Raw sensor readouts

- Customizable dashboard displaying real-time sensor data from engine controller
- Displays raw sensor readouts and converted values
- Control panel buttons send commands for remote valve actuation and ignition

Field Testing Setup



Future Plans

- Automated data transfer pipeline to extract test data from engine controller
- Enhanced post-test data analysis capabilities, including data visualization of processed data
- Support for additional sensors

Introduction

- Titan II started design in late Fall 2019, started manufacturing phase mid Fall 2020, and became test-ready in Fall 2021
- Implemented improved testing strategies, documentation, and technical upgrades to previous engine systems

Design, manufacture, and test a flight-optimized hybrid rocket engine to fly a custom airframe to 30,000 feet

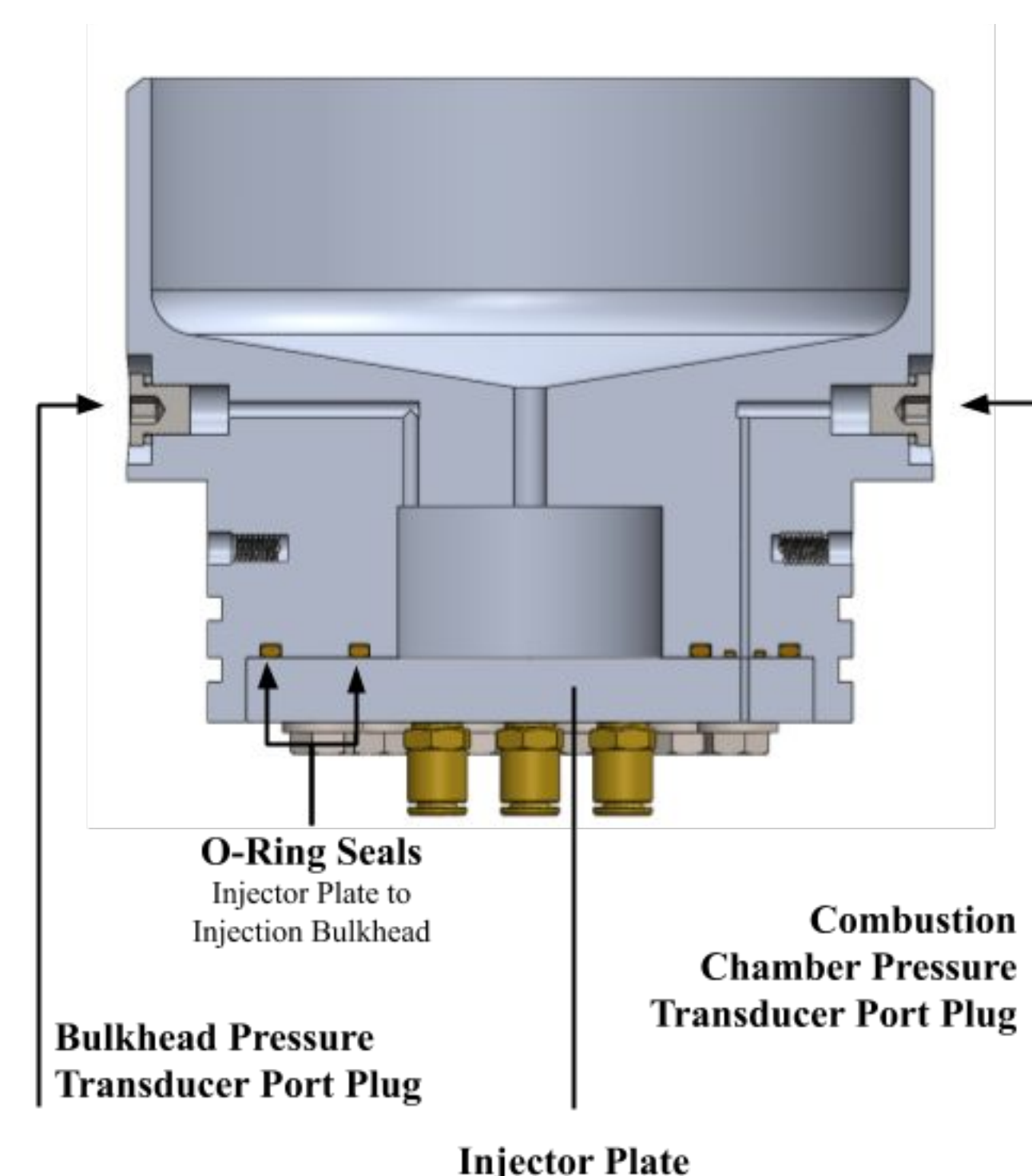
Engine Characteristics

Optimal Thrust	1200 lbf
Target Chamber Pressure	500 psi
Solid Fuel	Hydroxyl-Terminated Polybutadiene (HTPB)
Liquid Oxidizer	Nitrous Oxide
Max Burn Time	7.67 seconds

Sensor/Control System

Component	Quantity	Range	Sample-rate
Load cell	1	1769 lb	2 kHz
Thermocouple/probe	3	0-400 C	20 Hz
Pressure Transducer	4	0-1000 psig	500 Hz
Valves	6	N/A	N/A

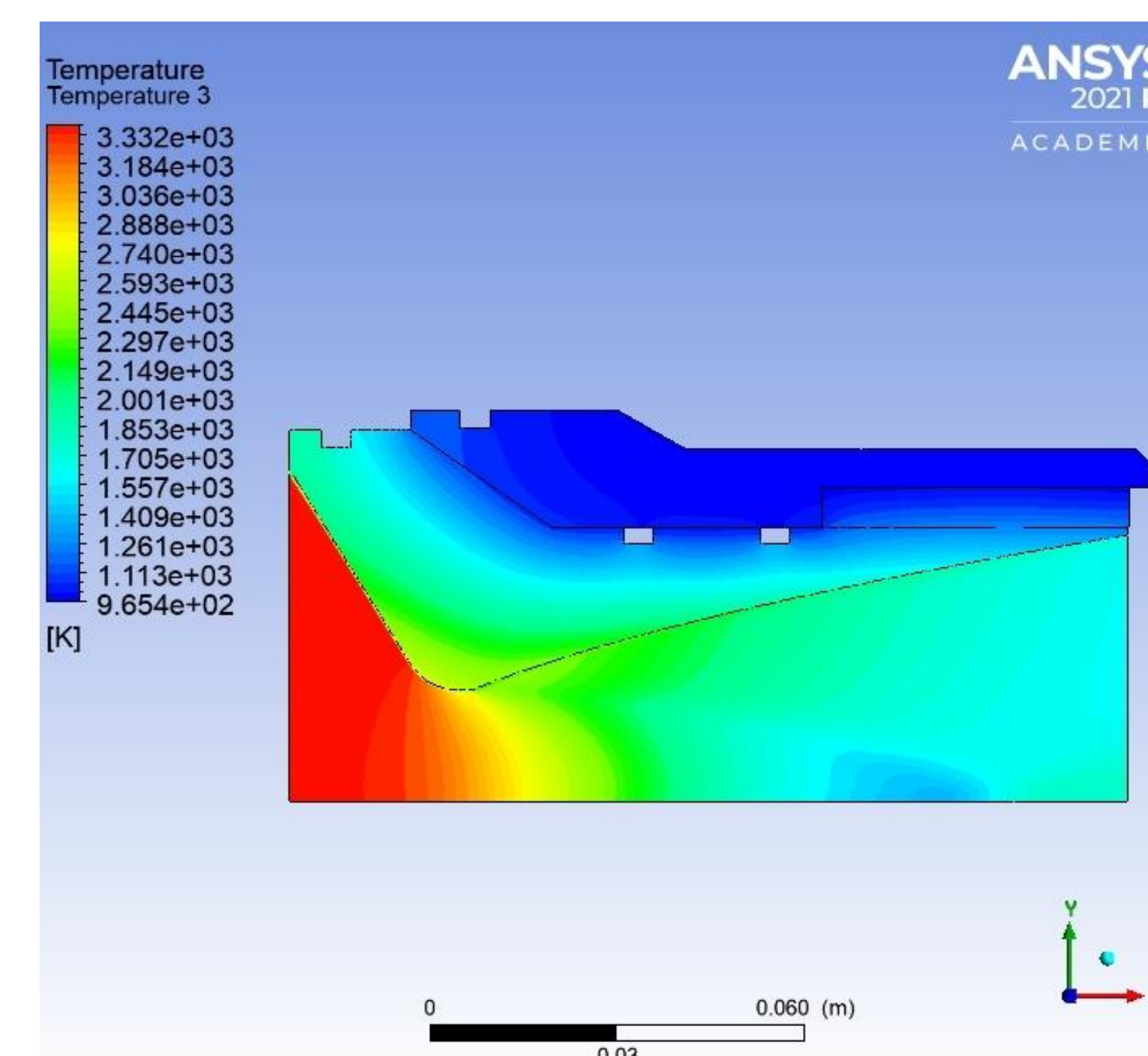
Design Highlights



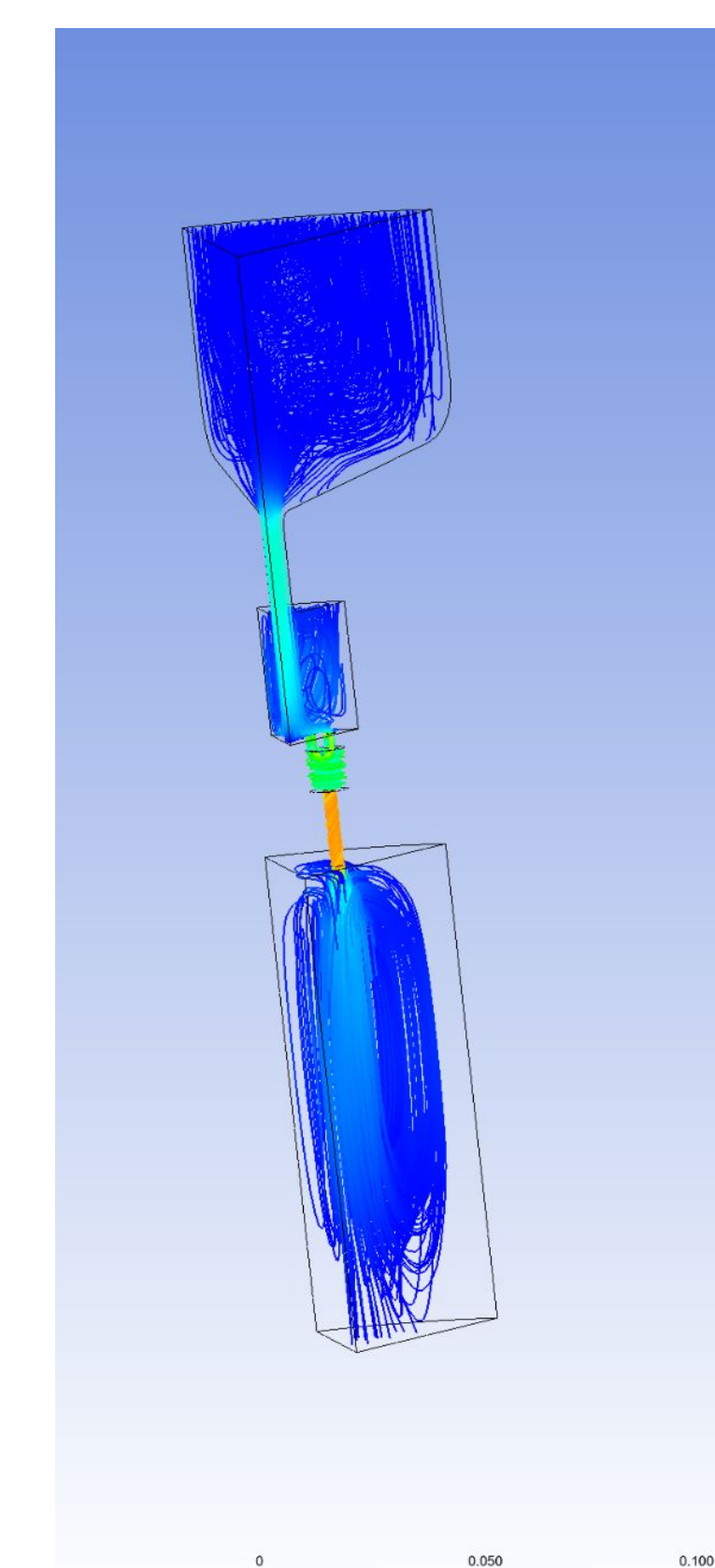
Cross-Section of Injection Bulkhead



Cross-Section of Rao Bell Nozzle

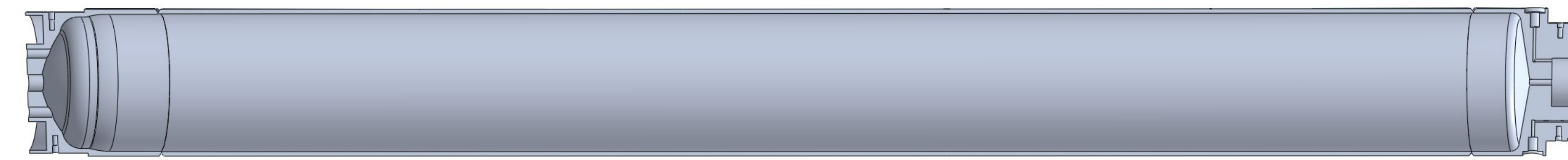


Thermal Flow Simulation of Nozzle Assembly

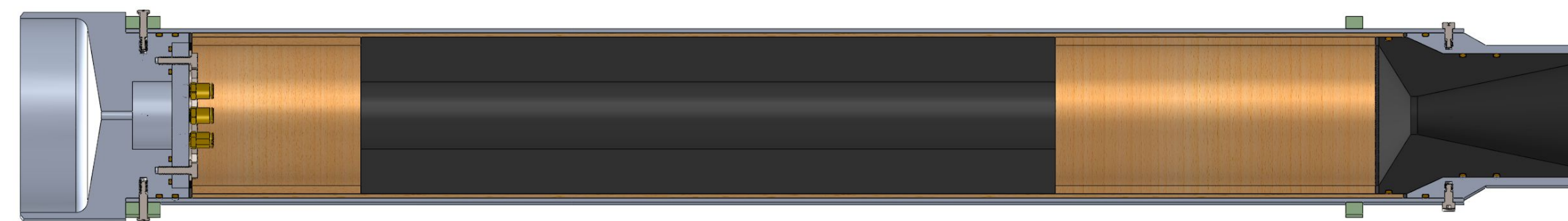


Injector Plate Fluid Velocity Simulation

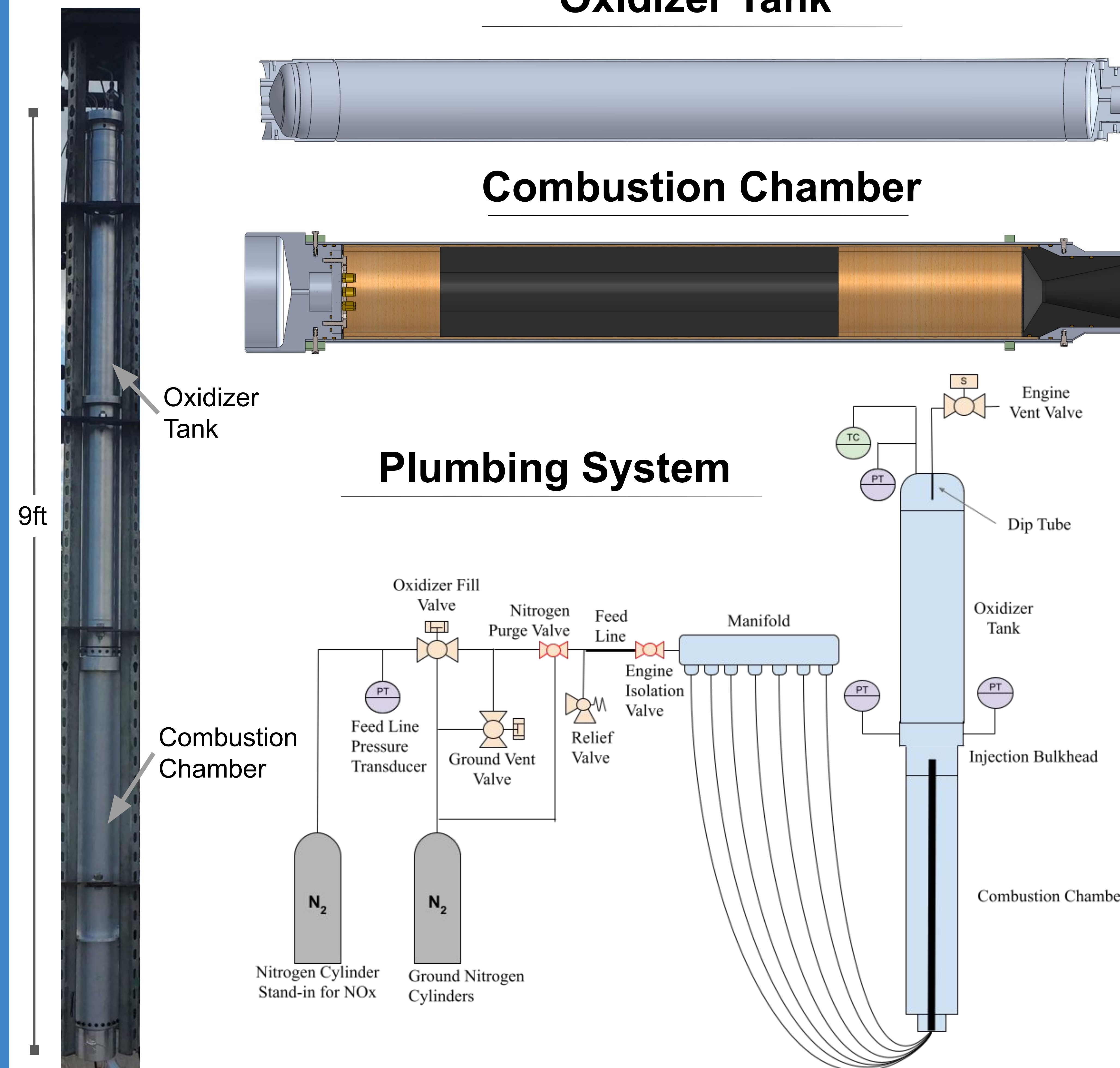
Oxidizer Tank



Combustion Chamber



Plumbing System



What is Eclipse?

Rice's rocketry team! We develop launch vehicles for intercollegiate rocketry competitions, rocket engines, avionics hardware & software!

Mobile Test Stand

- Vertical test stand more accurately simulates flight configuration
- Tower lowers for safe and easy transport to distant test site
- Cables anchor test stand to ground during test
- Flame diverter prevents exhaust from damaging equipment and surroundings
- Pneumatically-actuated valves control the flow and fail safe in case of electronics failure
- Pressure transducers and load cells record engine test data



Titan II integrated with Mobile Test Stand in Testing Configuration

Future Plans

- Static Hot Fire Titan II, 3 times
- Optimize Titan II w/ Test Data
- Integrate Titan II w/ Airframe
- Fly Archimedes

Special thanks to our corporate and university sponsors, and many other individual donors!



Introduction

- Hybrid rocket engines use a liquid oxidizer and a solid fuel grain
- Engine initially built last year but there were many ways to improve the engine this year
- Adaptable research tool meant to develop student skills
- Innovations and experiences from this engine will be applied to larger flight hybrid systems

Implement improved testing strategies and technical upgrades to previous hybrid rocket engine system

Engine Characteristics

Optimal Thrust	50 lbf
Target Chamber Pressure	500 psi
Solid Fuel	Hydroxyl-Terminated Polybutadiene
Liquid Oxidizer	Nitrous Oxide
Max Burn Time	10 seconds

Avionics System

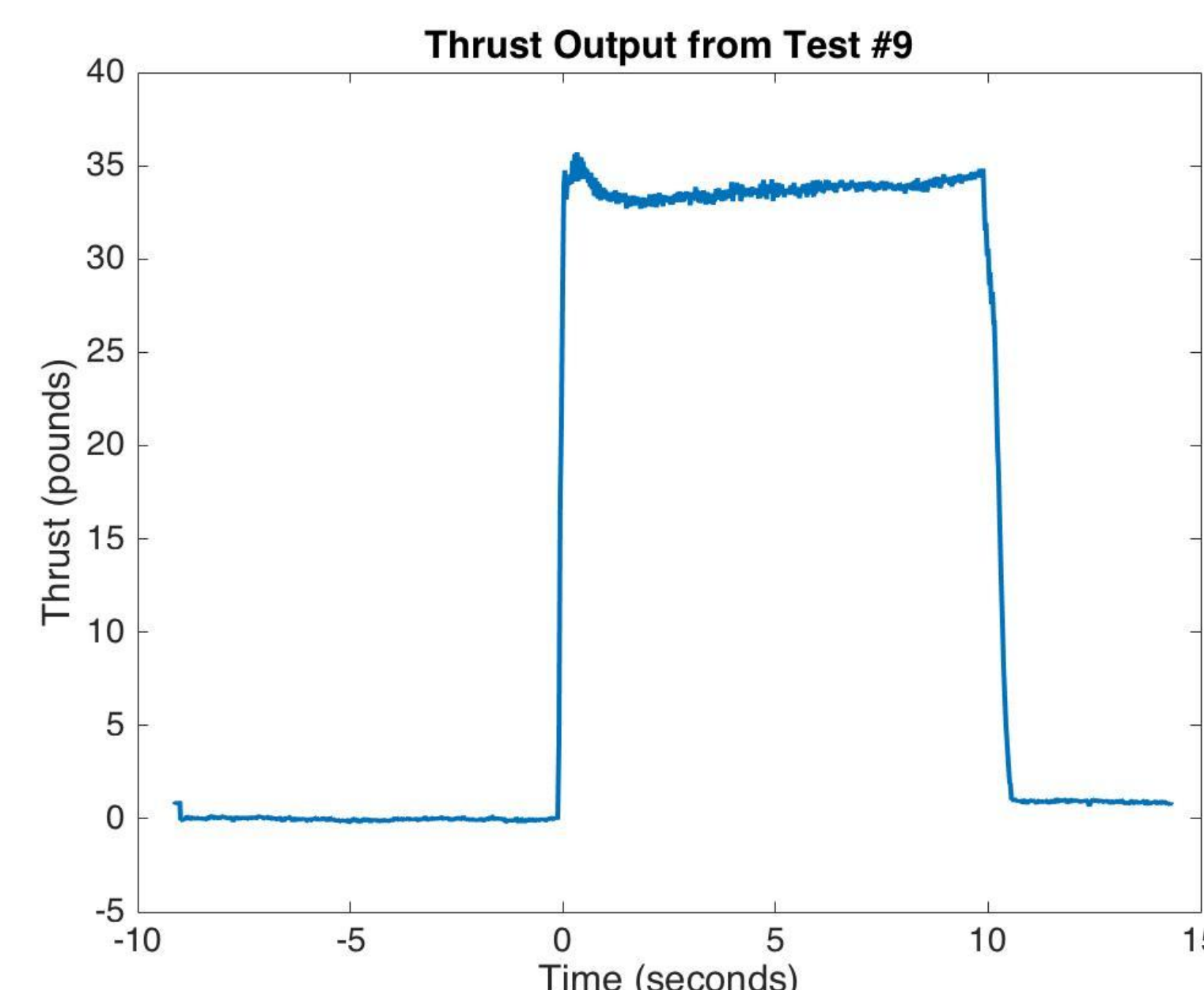
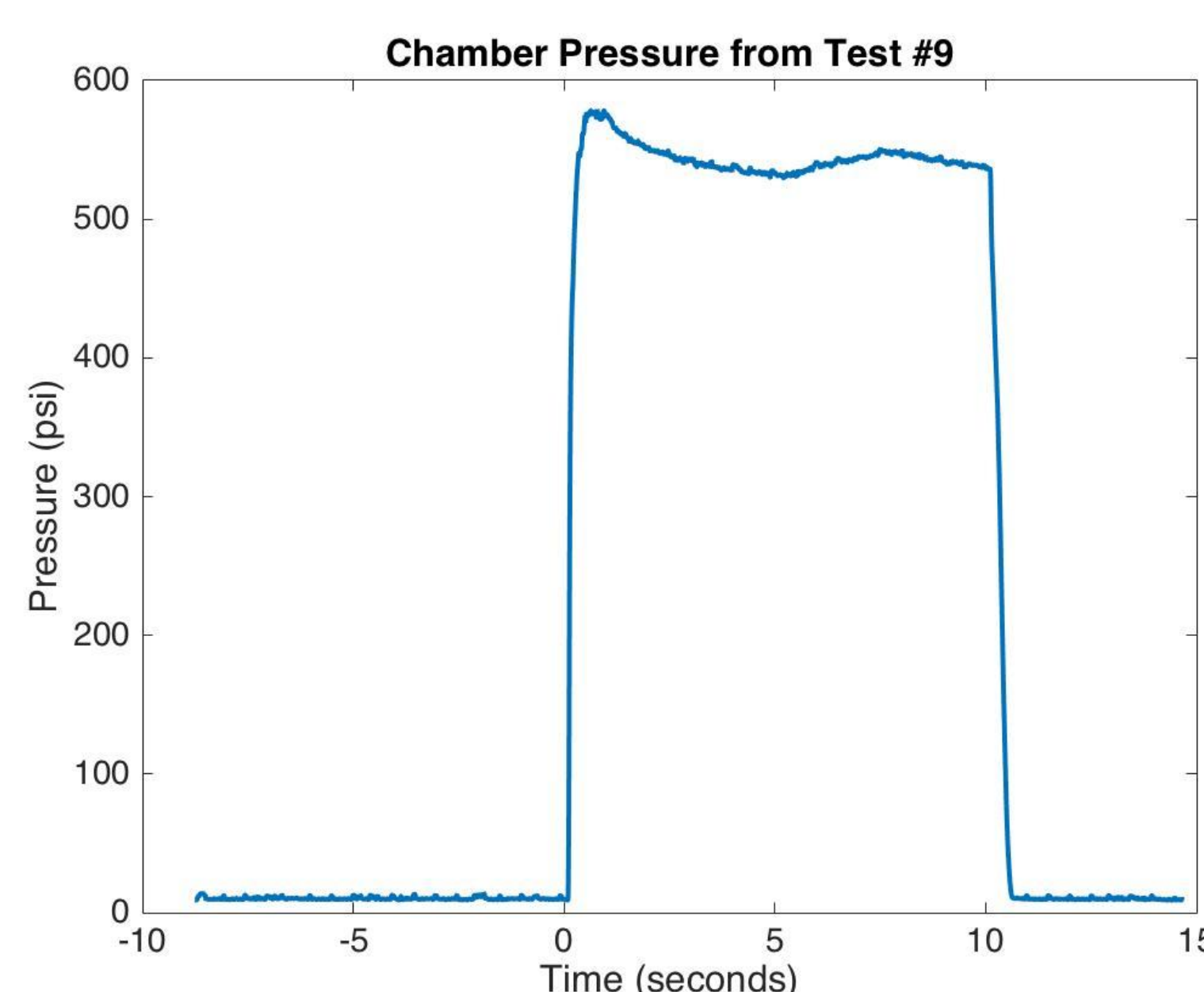
Component	Quantity	Range	Sample-rate
Load cell	4	0-100 lb	2 khz
Thermocouple	3	0-400 celsius	50hz
Pressure Transducer	3	0-1000 psi	1 khz
Valves	3	NA	NA

Testing

Test Number	Burn Length	Max Thrust	Chamber Pressure
4	10 sec	34 lbs	335 psi
5	9 sec	48 lbs	505 psi
6*	N/A	N/A	N/A
7	7 sec	35 lbs	450 psi
8	10 sec	35 lbs	540 psi
9	10 sec	37 lbs	550 psi

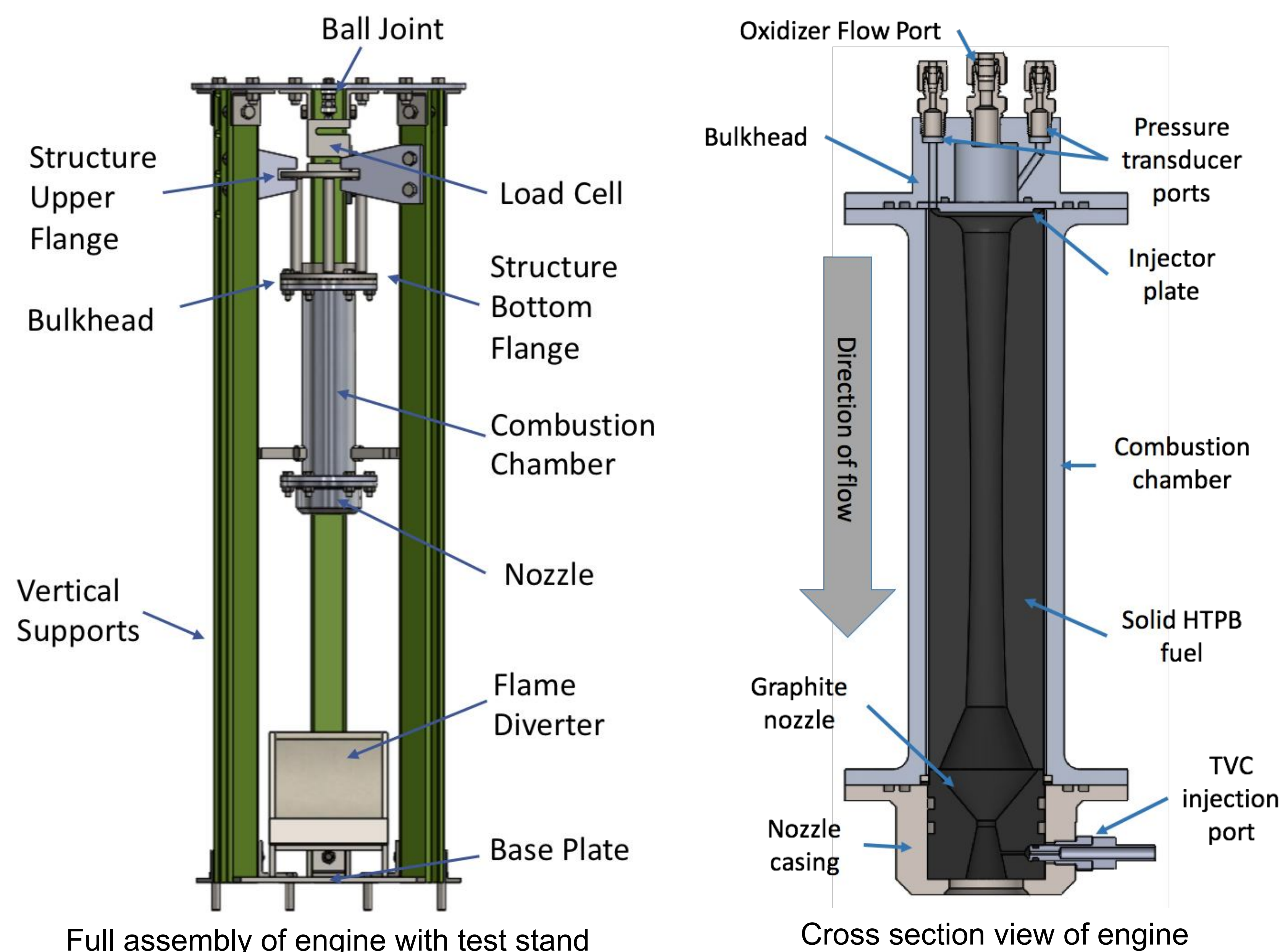
* Test #6 scrubbed due to insufficient fuel grain cure

- Each test applies small changes to the injector plate and fuel grain
- On March 3rd, Eclipse tested two separate engines (tests 8 and 9) on the same day
- In the future, having two engines will allow for significantly more testing and development of nozzles, injector plates, and fuel grains

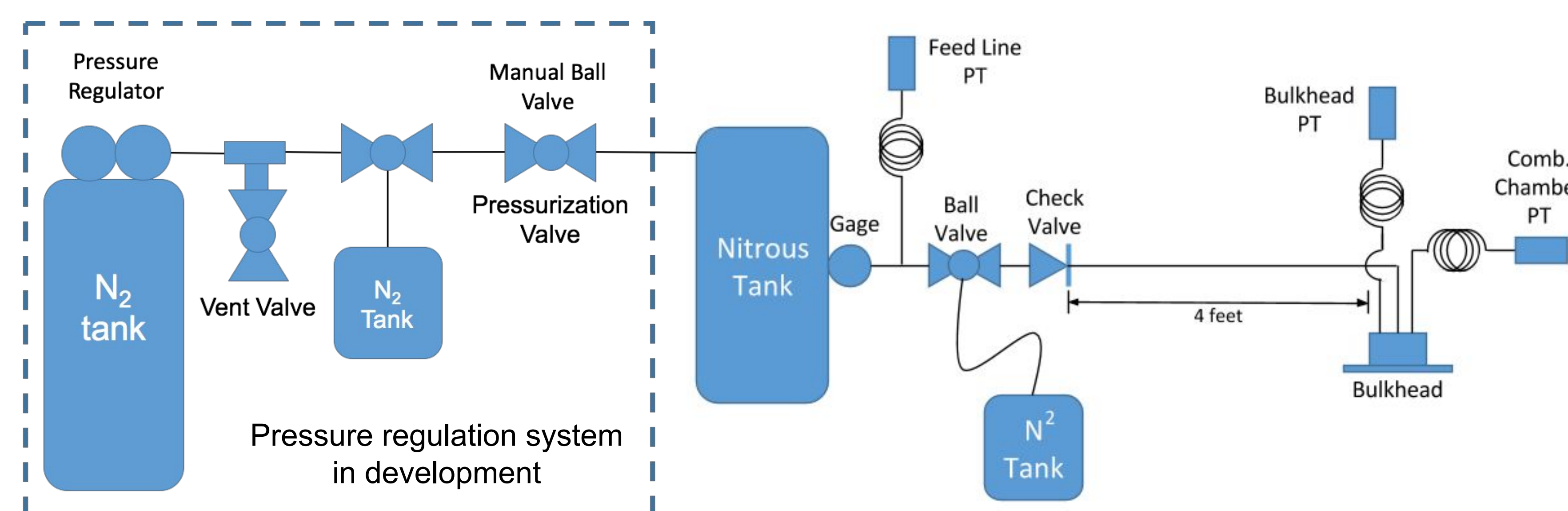


Still image from seventh engine test, Dec. 2018

Overall Engine Design

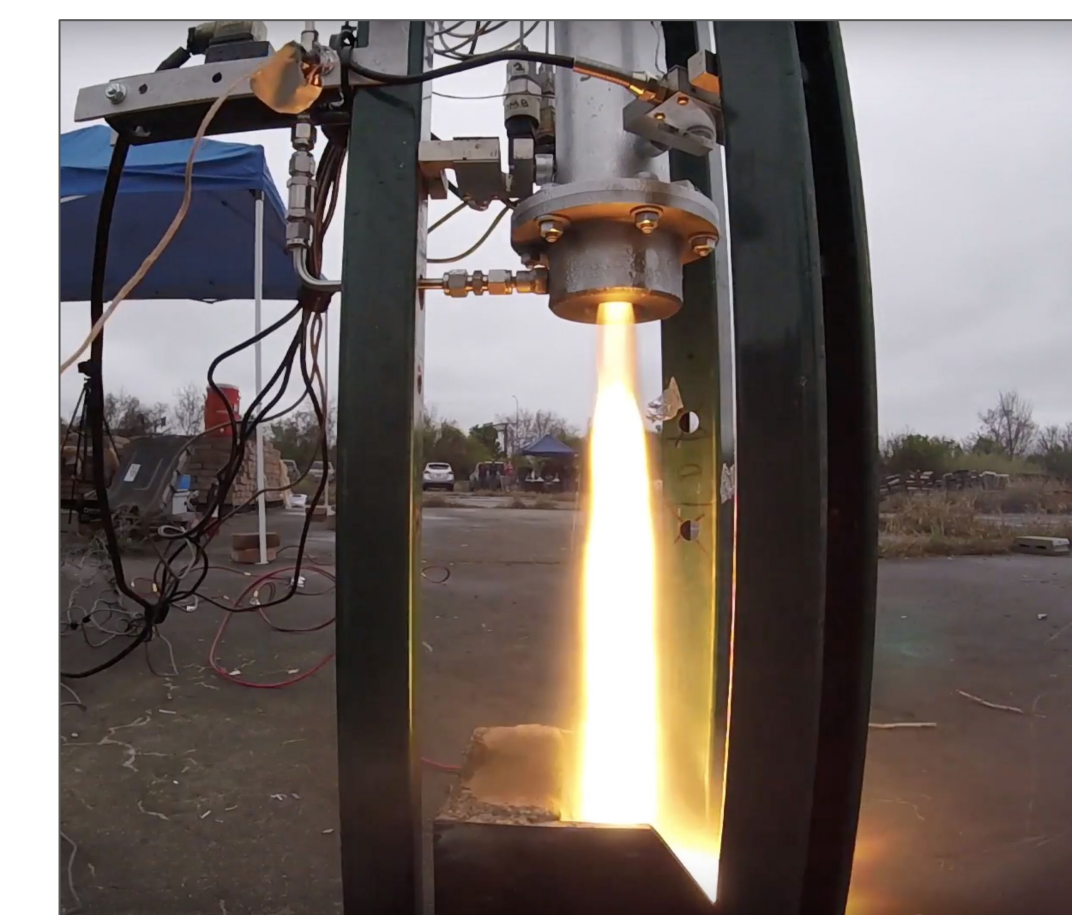


Plumbing System

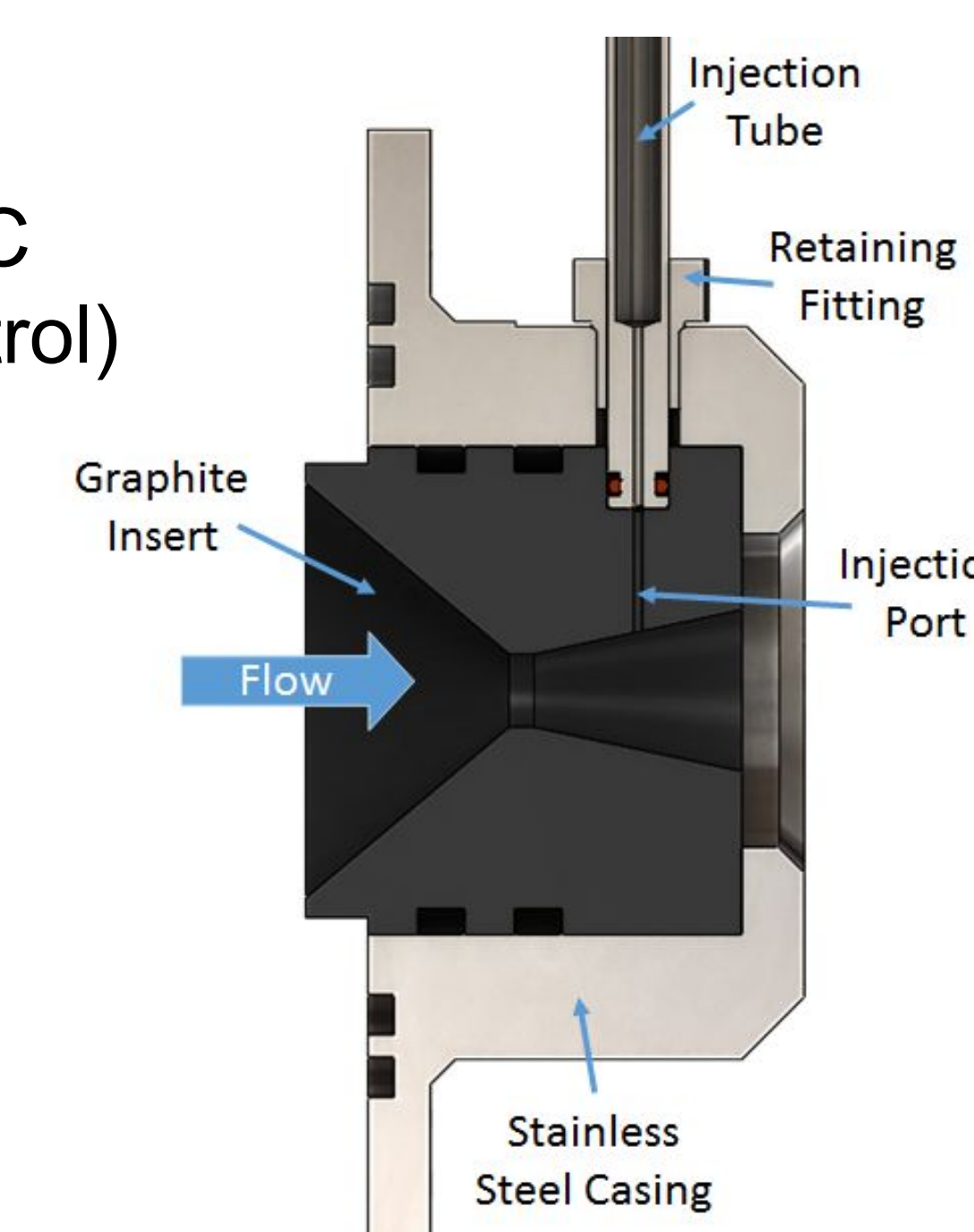


Thrust Vector Control

- Goal is to inject inert gas (N_2) into the diverging end of the nozzle
- Causes deflection of the exhaust velocity vector
- Successfully implemented GITVC (Gas Injected Thrust Vector Control) at 4 Luna tests



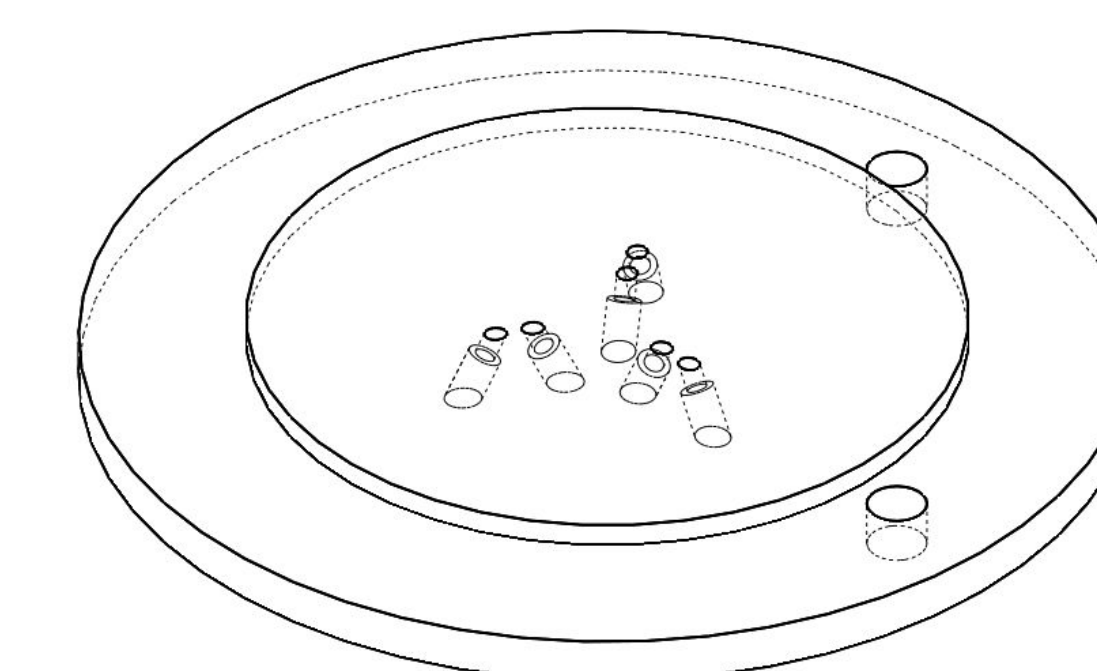
Still image from eighth engine test, March 2019



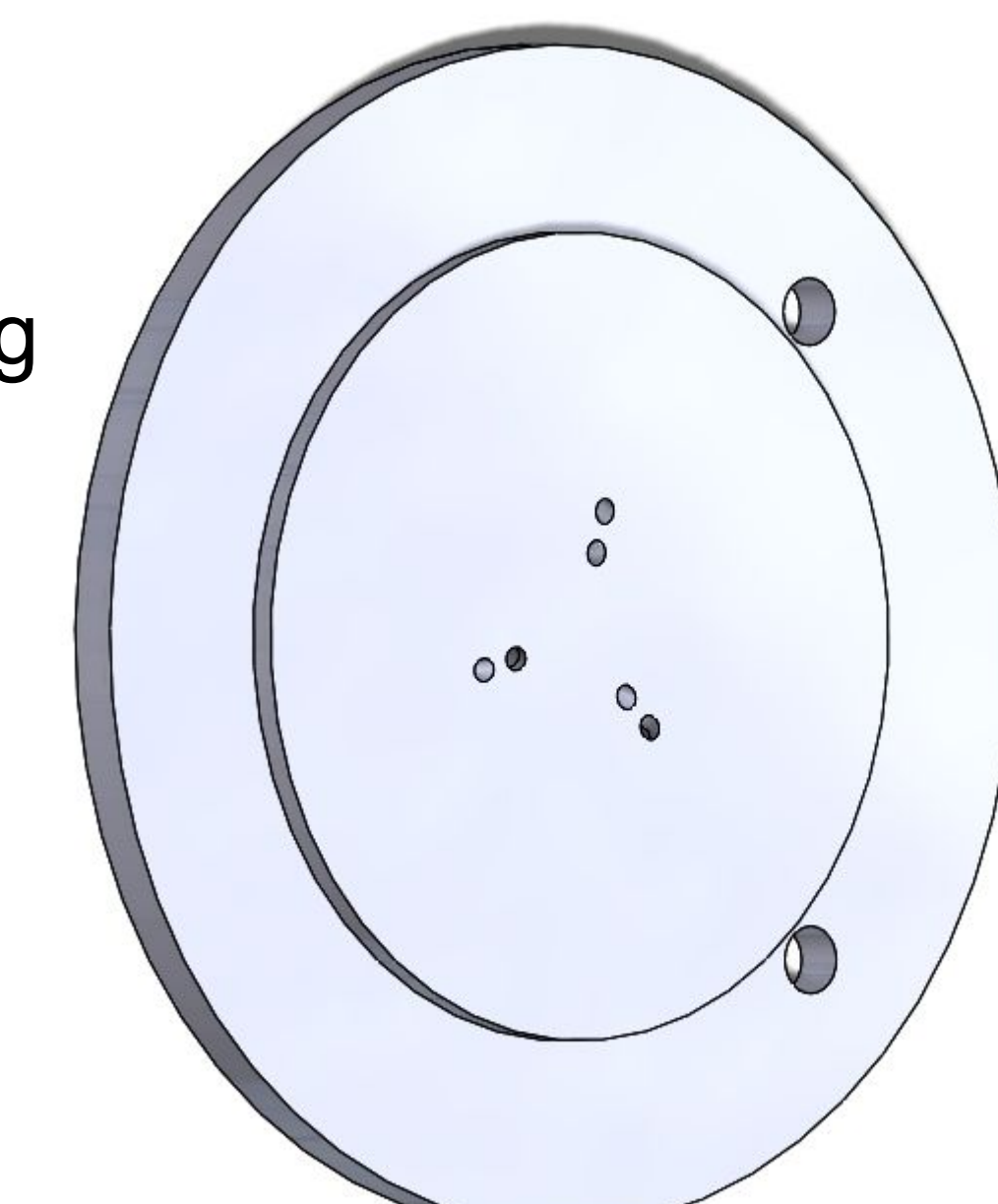
CAD Image of GITVC Nozzle

Impinging Injector Plate

- Designed so injected nitrous will collide and achieve atomization
- Three pairs of holes angled at 23 degrees
- Successfully implemented impinging injector plate during test number 8



Drawing of impinging plate, with visible ports



CAD Image of Impinging Plate

Future Plans

- Achieve full thrust by adjusting injector design
- Implement nitrous pressure regulation via N_2 gas injection into tank head space
- Bell shaped nozzle for increased thrust
- Flame diverter constructed of flame resistant concrete with integrated water cooling system

Special thanks to our corporate and university sponsors, and many other individual donors!



Introduction

- Develop launch vehicles, rocket engines, and avionics hardware & software for **international rocketry competitions**
- The team's long term goal is to build a rocket to fly our custom hybrid engine, Titan II

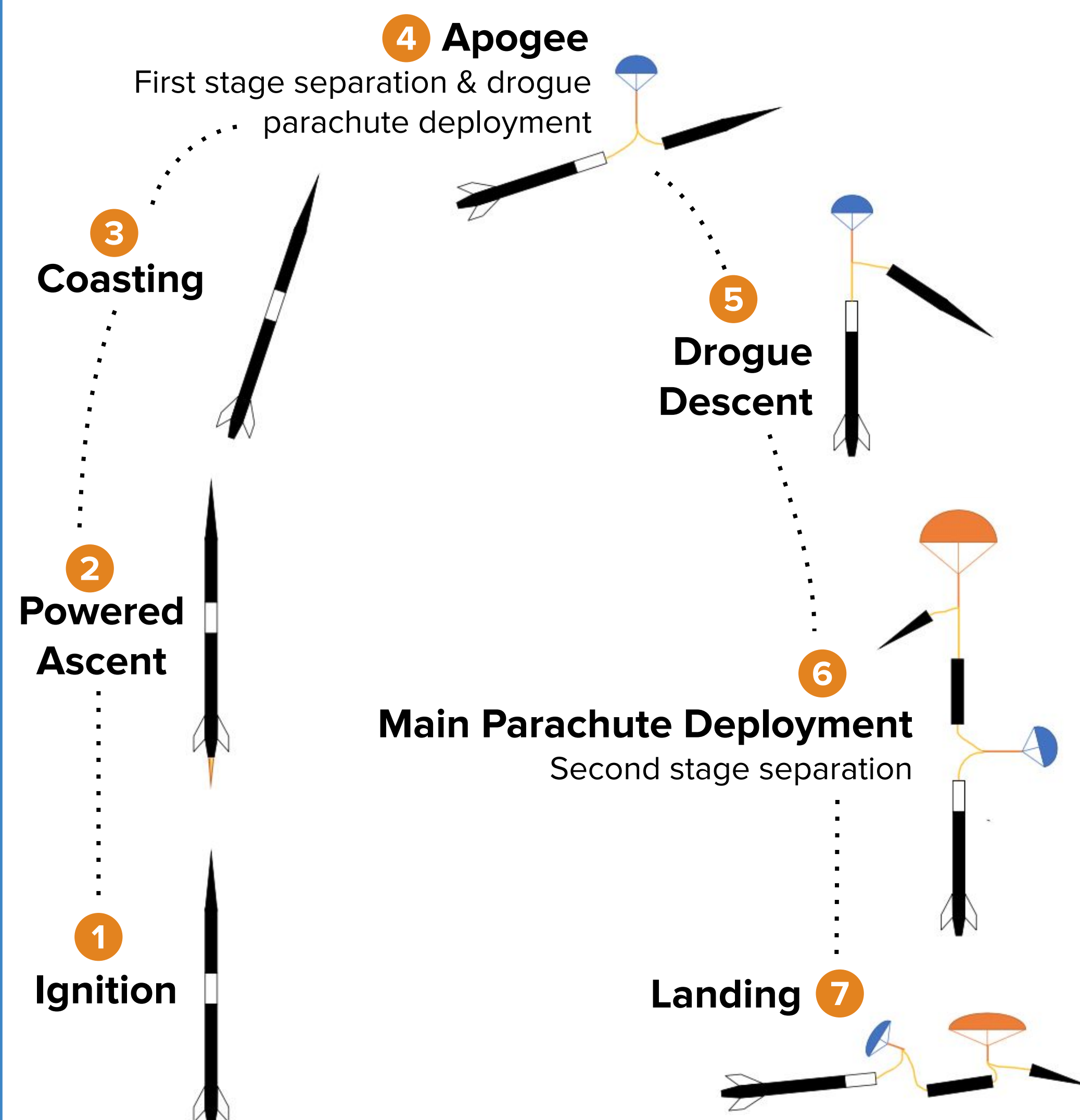
Project Goal

Design, build, and launch a **custom rocket airframe** to **30,000 feet** to demonstrate the system's ability to **safely integrate with a student-built engine**.

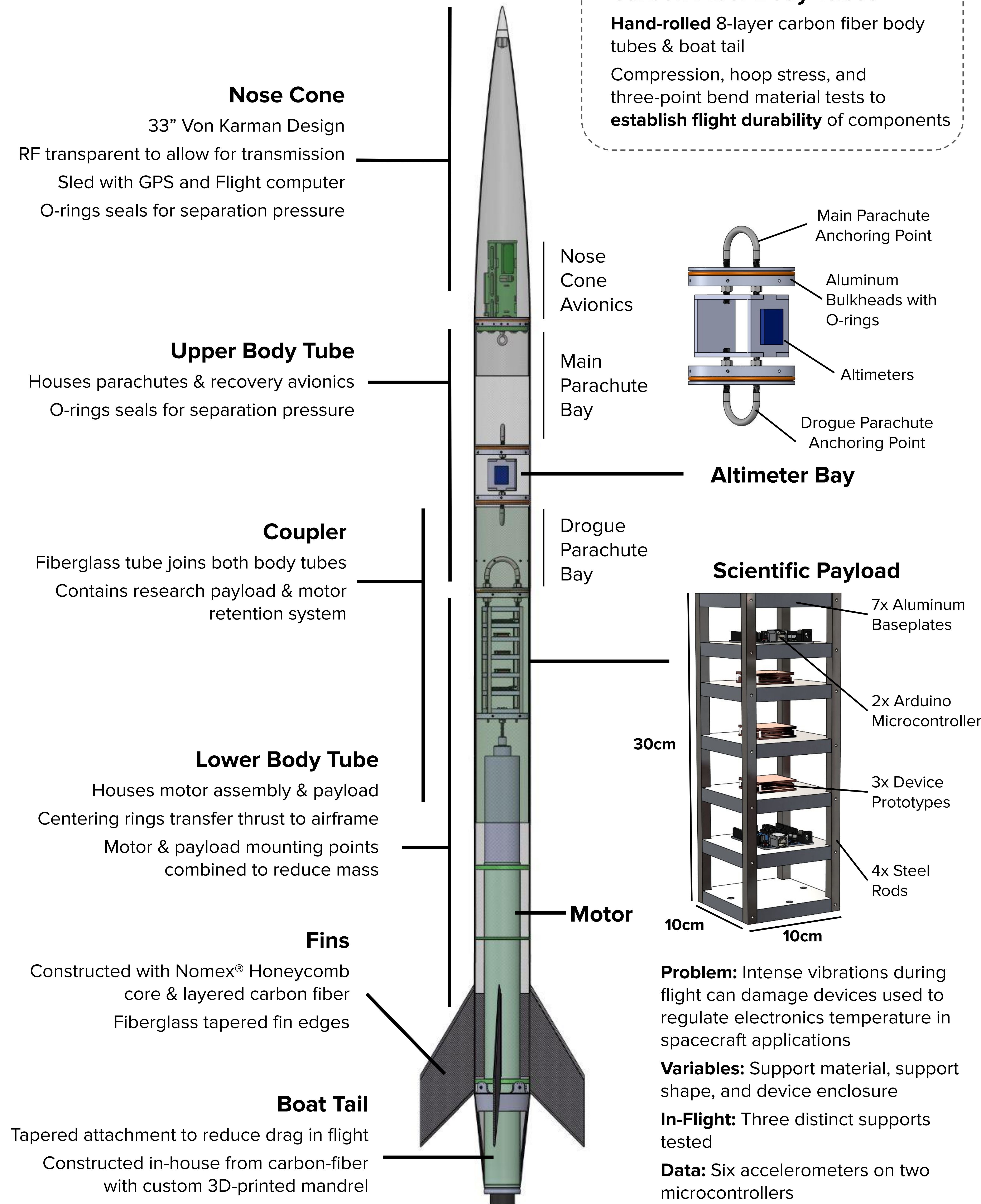
Minerva Specifications

Metric	Value
Rocket Length	130 in
Weight at Liftoff	92.7 lb
Motor Thrust	765 lbf
Peak Altitude	30,601 ft
Time to Apogee	45.1 s
Max Velocity	1710 ft/s (Mach 1.52)
Max Acceleration	407.65 ft/s ²

Phases of Flight

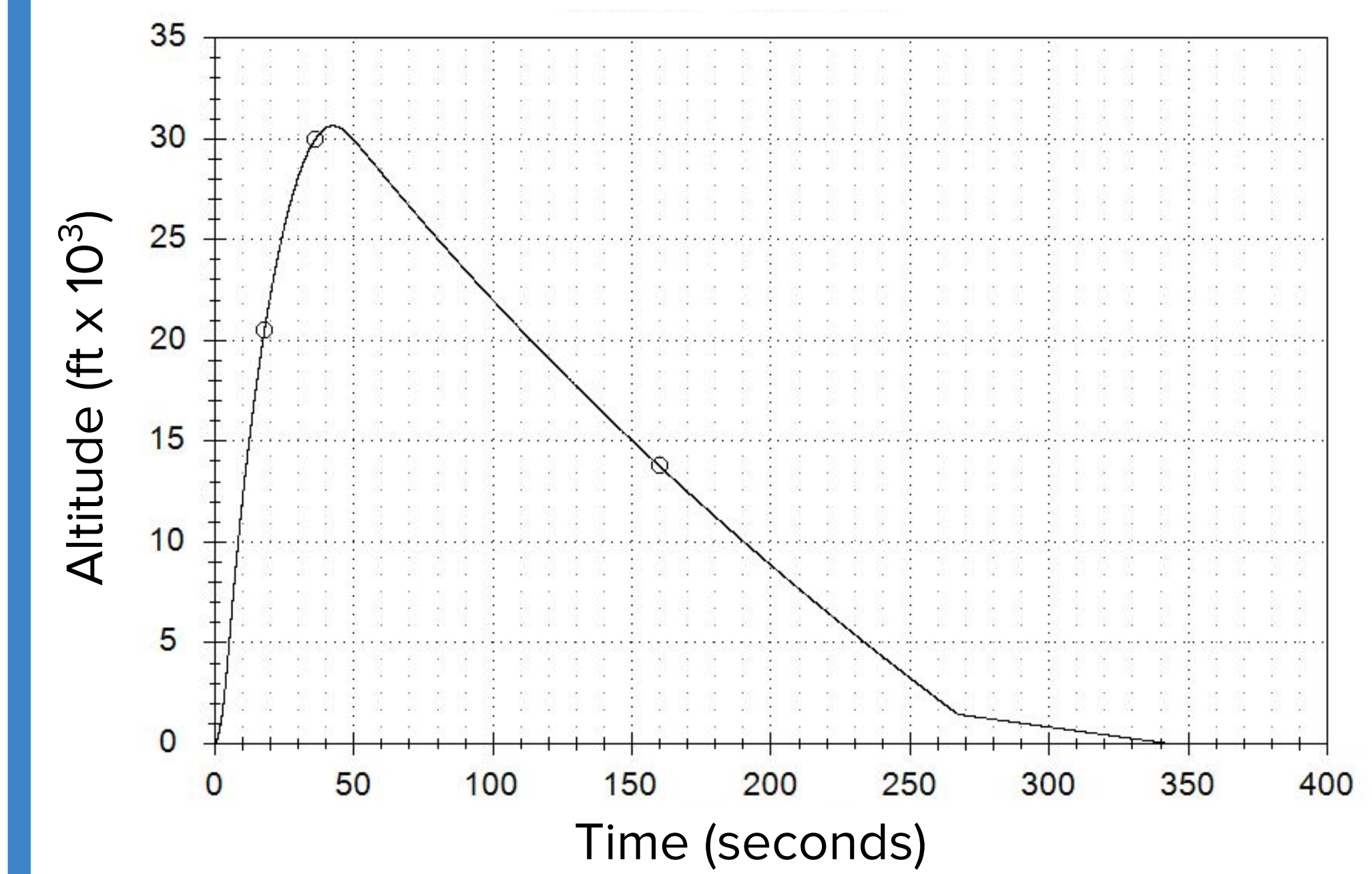


Full Minerva Assembly



Minerva Flight Altitude vs. Time

RASAero Simulation



Recovery System

- Dual-deployment system**
 - Apogee**: Upper body tube separates from coupler for drogue deployment
 - 1500 ft**: Nose cone separates from upper body tube for main deployment
- Barometric altimeters trigger black powder charges to **pressurize airtight chambers** and cause separation
- GPS transmits from nose cone so Minerva can be located
- Shock cords** hold all three sections together throughout descent
- Tested with **2,500 ft prototype rocket**



Successful Minerva Separation Test (04/10/2022)

Future Plans

- Test launch of Noctua III on April 23rd
 - Collect altitude and GPS data
 - Collect data on RTR flight computer
- Compete in New Mexico on June 21st**
- Integrate with Titan II next year

Special thanks to our corporate and university sponsors, and many other individual donors!



Introduction

- Rice Eclipse is competing in the Spaceport America Cup in the 30,000 ft apogee COTS Solid Motor Category
- Real-Time Rocket (RTR) is an iterative design process, with Electro II being the 3rd stage
- Testing an entirely student-built board with custom designed peripherals and power system

Project Goal

Create a **reusable, cost-effective, and modular** flight computer to **collect and transmit sensor data** from Eclipse rockets.

Sensors

- Sensors are designed to be standardized, modular and “plug-and-play”
- Board has support for 2 I²C sensors, 2 SPI sensors, and 1 UART sensor

Altimeter

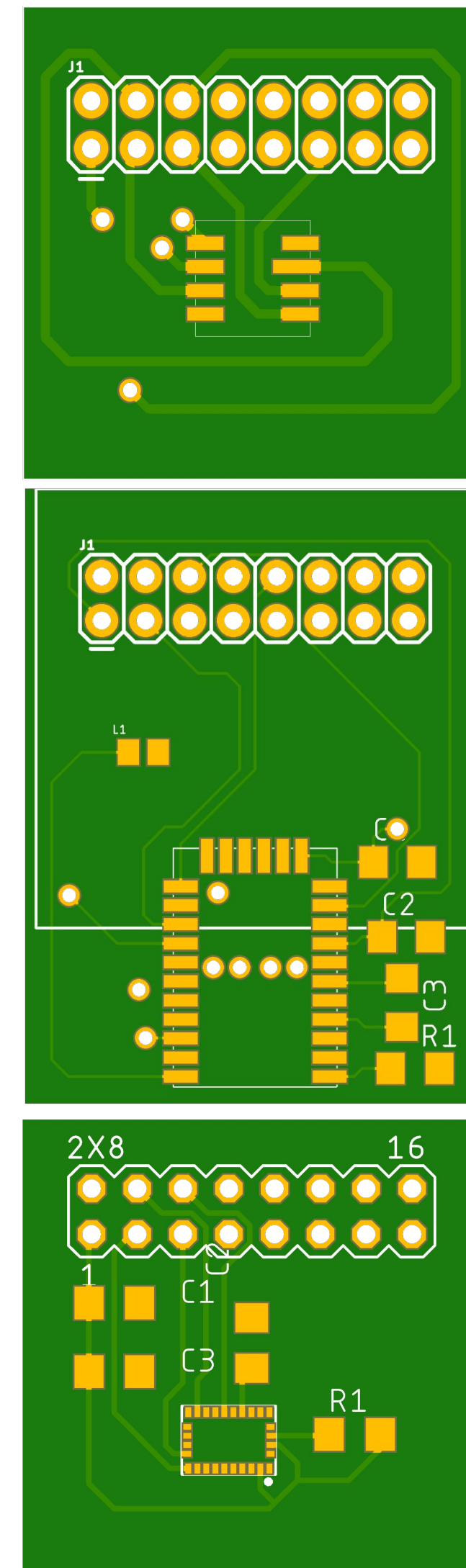
- TE Connectivity MS5803-01BA communicating over I²C
- Altimeter measures the rocket's apogee and its vertical trajectory

GPS

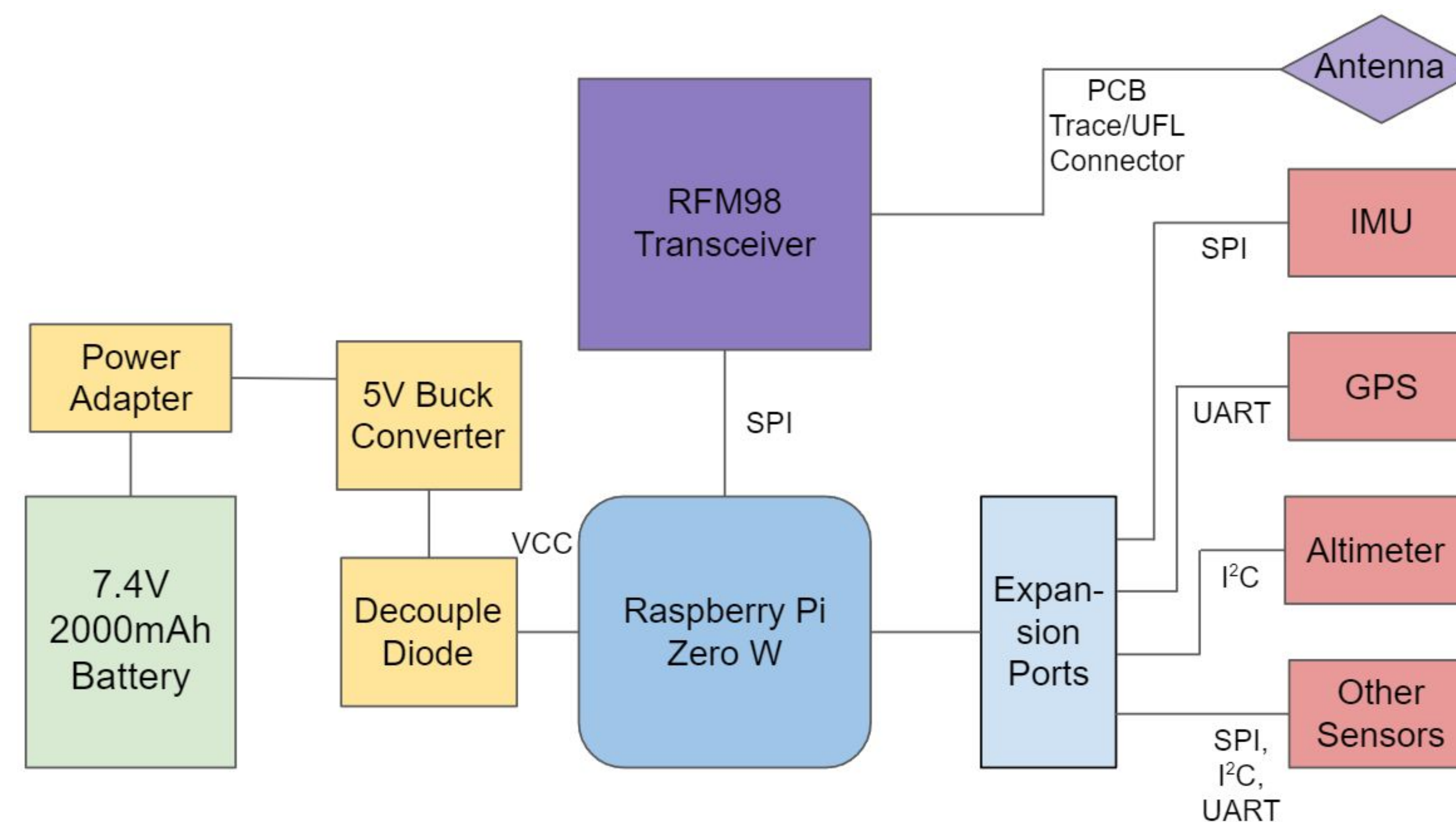
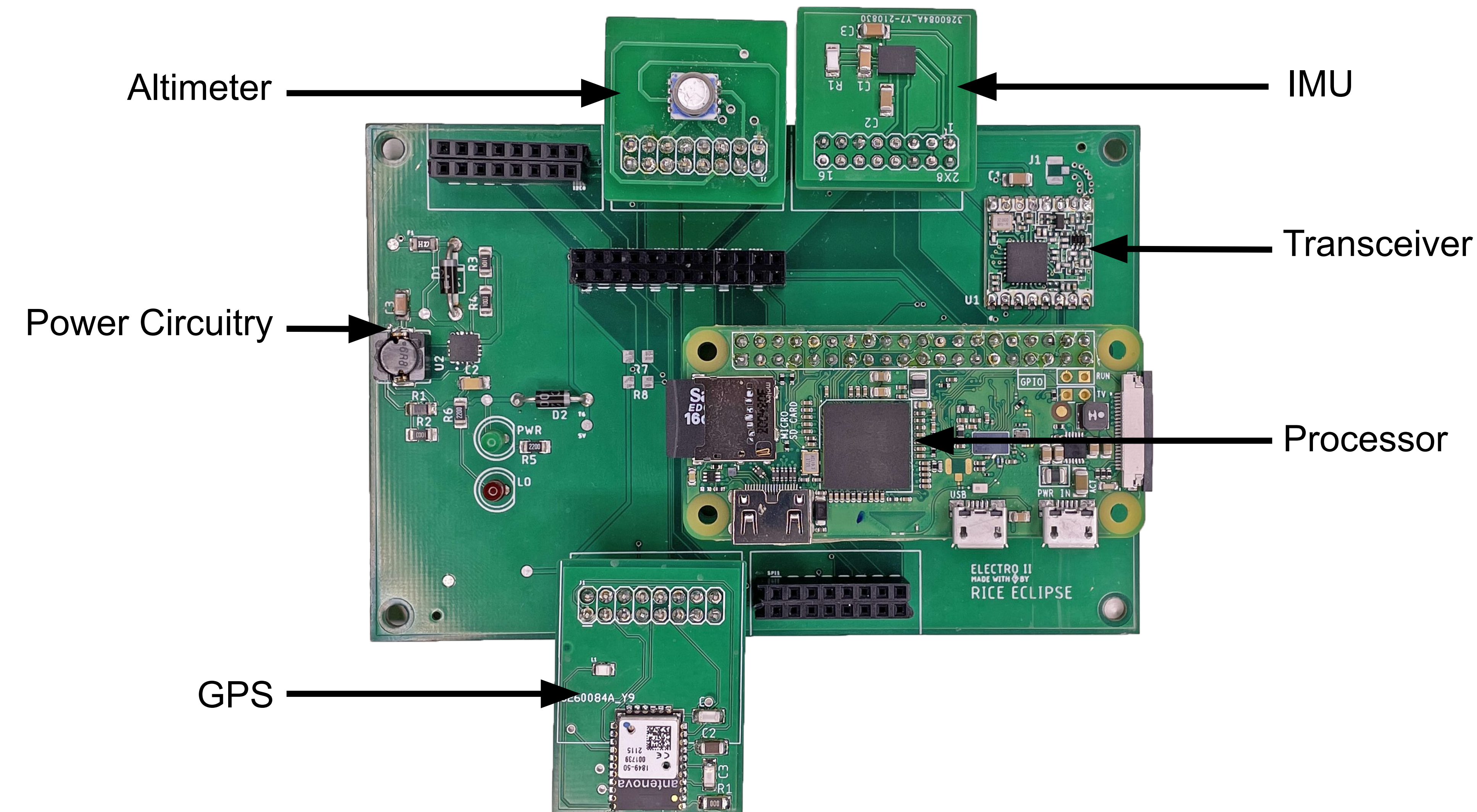
- Antenova M20050-1 communicating over UART
- GPS tracks the rocket for recovery and measure its horizontal trajectory

IMU

- Hillcrest Labs BNO055 9-axis IMU communicating over SPI
- Triaxial accelerometer, gyroscope, and geomagnetic sensor get orientation data



Overall Design

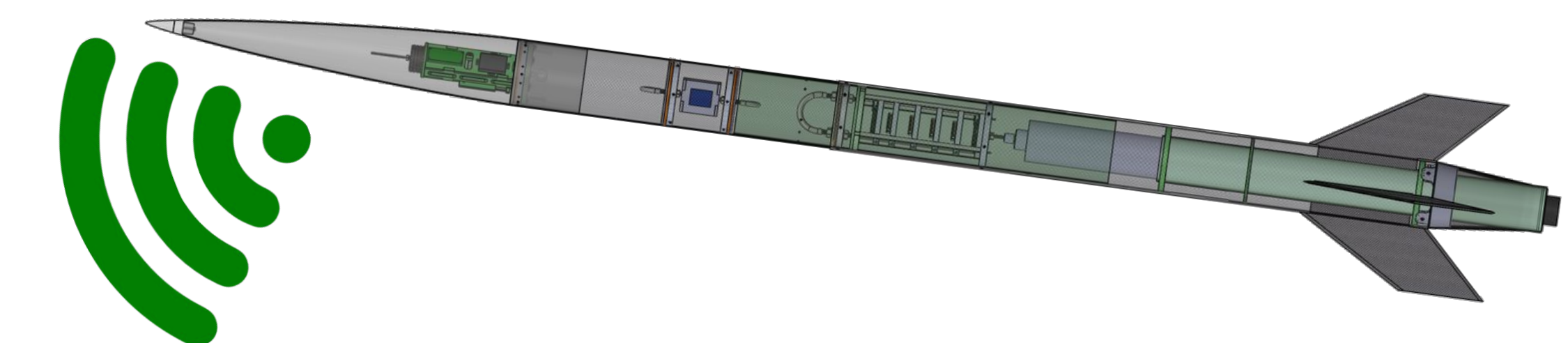


Design Parameters

Parameter	Desired Value
Maximum Transmission Altitude	≥ 30,000 ft
Battery Life	≥ 3 hours
Sensors Supported	≥ 4
Board Size	6 in x 4 in
Sensor Breakout Size	1 in x 1 in
Communication Frequency	433 MHz

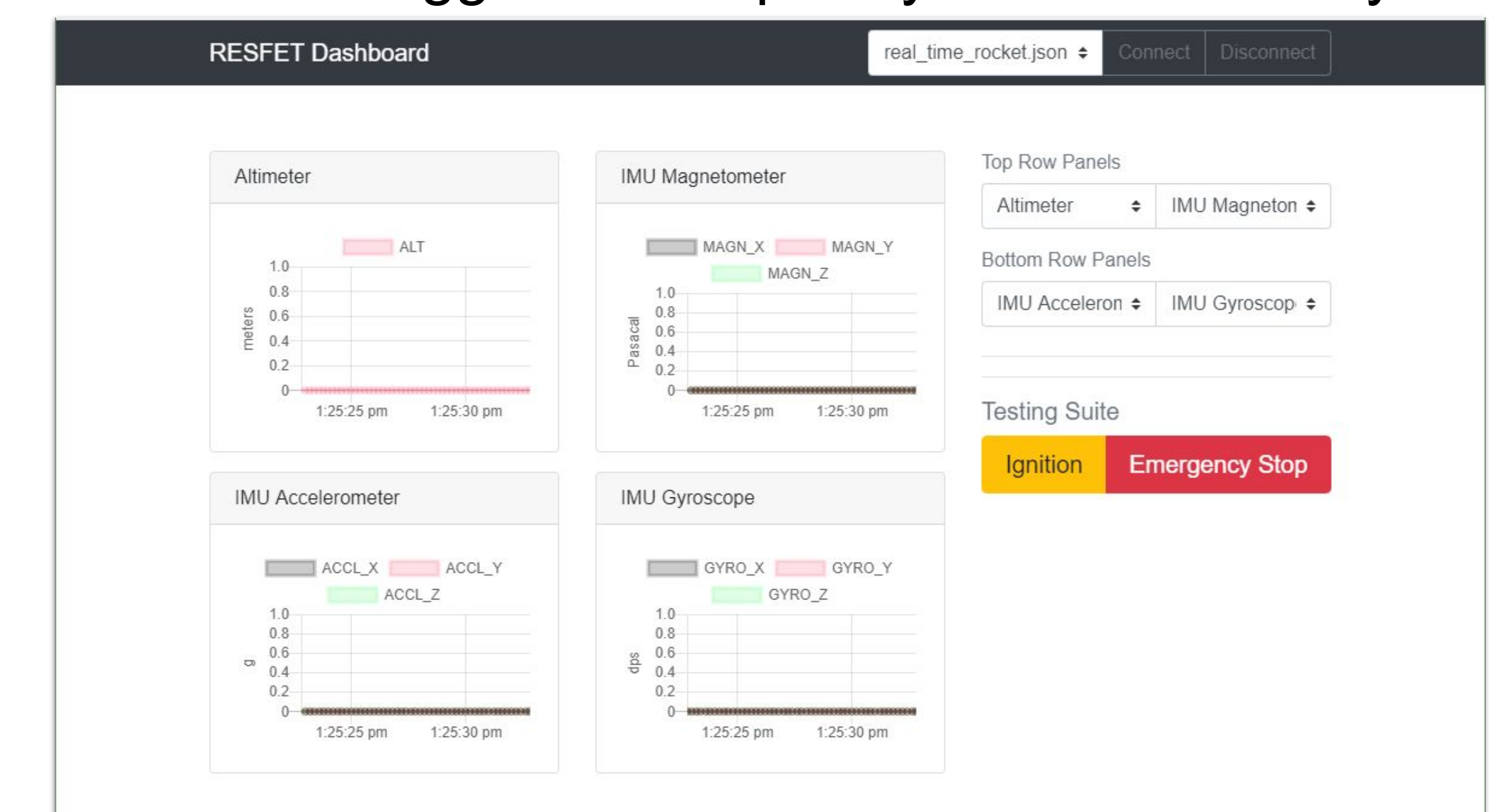
Communication

- Requires a range of 30,000 ft to communicate with RTR in flight
- Built in-house as a BiQuad antenna designed to achieve a gain of approximately 10 dBi
- RFM98 transceiver operating at 433 MHz
- 433 MHz requires HAM-certification, guaranteed to be less congested than typical 915 MHz band



Mission Control

- Data received by the ground transceiver is relayed to an Arduino which outputs serial data to a mission control laptop
- Visualization software (Electron app written in Javascript) plots/logs incoming data real-time
- Data also logged to Raspberry Pi for later analysis



Power

Component	Ave. Current	Max. Current
Raspberry Pi	150 mA	250 mA
GPS	21 mA	29 mA
Altimeter	<1 mA	1 mA
IMU	1 mA	15 mA
Transceiver	10 mA	120 mA
LEDs (x2)	40 mA	40 mA
Total	222	455

- 7.4V 2000mAh battery for a minimum of 4 hours of flight time
- Steps down voltage for 5V with buck-converter
- Varied trace sizes for expected power throughput to avoid overheating
- Ground plane across entire PCB to avoid bottleneck and heat risk

Future Plans

1. Improve GPS and IMU accuracy

2. Compete in Spaceport in June

3. Antenna: 30000 ft communication

4. Neutro: high fidelity redesign

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